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# Unit 2 Micronutrients - Provide me revision papers of FQE

nutrition and dietetics (Kenya Medical Training College)

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# Micronutrients- Vitamins and Minerals;

## MINERALS

### Definition of terms

- **Major minerals**:

These are essential mineral nutrients required in the adult diet in amounts greater than 100 milligrams per day. *calcium, phosphorus, potassium, sodium, chloride, magnesium and sulfur.*

- **Trace minerals:**

Essential mineral nutrients required in the adult diet in amounts less than 100 milligrams per day. include *iron, copper, cobalt, zinc, manganese, iodine, molybdenum, fluorine, selenium and chromium.*

- **Fluid and electrolyte balance:**
  Maintenance of the necessary amounts and types of fluid and minerals in each compartment of the body fluids.
- **Acids:** compounds that release hydrogen ions in solution.
- **Bases:** compounds that accept hydrogen ions in solution.
- **Acid- base balance**: The balance maintained between acid and bases concentrations in the blood and body fluids.
- **pH:** The concentration of hydrogen ions. The lower the pH, the stronger the acid. Thus pH 2 is a strong acid while pH 6 is a weak acid; Ph 7 is neutral; and a pH above 7 is alkaline.
- **Buffers**: Compounds that can reversibly combine with hydrogen ions to help keep solutions acidity constant.
- **Ossification**: The process of bone formation.
- **Osteomalacia**: A weakening of the bones due to a calcium, phosphorus, and/or Vitamin D deficiency.
- **Osteoporosis**: An abnormal porousness of bone as the result of a



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calcium, phosphorus, and/or vitamin D deficiency.

**Classification**

Minerals are classified into two groups based on the amount found in the body. Some occur in large quantities and are required in quantities above 100mgs per day. These are referred to as macro-minerals and they include calcium, phosphorus, potassium, sodium, chloride, magnesium and sulfur. Others occur in small amounts and are required in only a few milligrams (15mgs/day or less). They are therefore considered as trace elements or micro-minerals and they include iron, copper, cobalt, zinc, manganese, iodine, molybdenum, fluorine, selenium and chromium.

**General Functions of Minerals include:**

Minerals perform several general functions because of their interrelationship. They provide structure to the body tissues and regulate body processes such as fluid balance, acid-balance, nerve transmission, muscle contraction, vitamin, enzyme and hormonal activities.

**a. Structure**

Calcium, phosphorus, magnesium and fluoride provide structure to bones and teeth. Soft tissues gain structural support from phosphorus, potassium, iron and Sulphur. Sulphur is also a fundamental constituent of skin, hair and nails.

**b. Maintenance of acid-based balance**

This term refers to the maintenance of the body's concentration of hydrogen ions (pH). Biological reactions in the cell occur when the pH is right. Anything that changes the pH of the cell environment may change the way cellular enzymes act. Inactivation of enzymes results in starvation and death of cells. Acid forming minerals such as, chlorine, Sulphur and phosphorus are found in the extra cellular fluid. They predominate in foods rich in protein such as meat, fish, poultry, eggs and cereal products (acid forming foods). Basic or alkaline minerals such as calcium, sodium,



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potassium and magnesium tend to predominate in fruits, vegetables and nuts. Milk contains internal balance of base forming calcium and acid forming phosphorus, and does not influence acid-base balance

**c. Catalysts for biological reactions**

Minerals serve as co-factors for many biological reactions. They are not part of initial compounds or end products but they serve as vital links between enzymes and the substance which they act on e.g. zinc in known to catalyze about 100 different reactions. They catalyze many of the separate steps involved in the metabolism of carbohydrates, fats and proteins. For example, the synthesis of haemoglobin depends on several mineral elements that are not part of haemoglobin such as iron. Similarly, although calcium is a catalyst for blood clotting, it is not part of the clot. The absorption of nutrients from the gastrointestinal tract and the uptake of nutrients by the cells often depend on minerals (e.g. calcium facilitates the absorption of $B_{12}$, sodium and magnesium facilitates absorption of carbohydrates).

**d. Components of essential body compounds**

Many hormones, enzymes and other vital compounds synthesized in the body contain minerals as essential parts of their structure. As such, if the required mineral is absent, the body will be unable to produce adequate amounts of these substances. For example, thyroxin, a hormone which controls energy metabolism requires iodine. Haemoglobin which transports oxygen to the cells and carbon dioxide from them contains iron; chlorine is essential for hydrochloric acid production in the stomach. Mineral containing enzymes are sometimes called metalloenzymes. If the mineral is unavailable, the enzymes will not be synthesized or will be ineffective. Molybdenum is part of an enzyme needed to release liver stores of iron. Zinc is part of a protein splitting enzyme carboxypeptidases secreted into the pancreatic juice.

**e. Maintenance of fluid balance**



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Water comprise of 60% total body weight or 70% of fat free weight. It is present in 3 components, separated by semi-permeable membrane that allows free exchange of fluids and electrolytes (ions). Intravascular (in blood) compartment includes fluid in the arteries, veins and capillaries. Intercellular (between cells) bathes individual cells and provides nourishment, this act as a buffer area. Its volume will adjust to prevent changes in the volume of the intravascular or extra cellular chambers. Intracellular fluid is found within the cells. Transcellular compartment includes the fluid which lubricates the joints and fluid which in the eyeball. This represents a very small portion and it is not involved in fluid shifts.

The movement of fluid from one compartment to another is controlled by the concentration of minerals on either side of the membranes. Mineral elements in body fluids occur as salts that separate into component ions with positive and negative charges. Charged ions are known as electrolytes and maintain osmotic pressure of body fluids. As the concentration of electrolytes increase, osmotic pressure increases, this draws fluid from the compartment with low electrolyte concentration until the concentration (or pressure) is equalized. Under normal circumstances, the body is able to prevent a shift in electrolyte concentration between compartments. Shifts however do occur in abnormal situations.

**f. Transmission of nerve impulses**

Minerals play a vital role in conducting nerve impulses along a nerve fibre. The exchange of sodium and potassium across nerve causes the transmission of nerve impulses. The permeability of membrane changes allowing potassium to leave the cell. At the same time sodium enters each cell, but is quickly pumped out, this creates a temporary change in the electrical charge. In this way the message is passed along the fibre. The transmission of a nerve impulse from one nerve cell to another is also dependent on neurotransmitter acetylcholine (ACH) at the junction of the two fibres. The release of ACH is regulated by calcium.



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g. **Regulation of muscle contraction**:

The muscles of the body are bathed in the intercellular fluid. For muscles to function normally, the composition of this fluid should have a balance between mineral elements such as calcium which stimulates muscular contraction and sodium, potassium, and magnesium which exerts relaxing effect. This balance does not depend on dietary sources and can only be disturbed by hormonal changes.

**h. Growth of body tissue**

Some minerals such as calcium and phosphorus occur in large concentration in bones and teeth and are the building material of body tissue. Absence of these will lead to stunted growth or poorly developed bones and teeth. Other minerals serve as catalysts to reactions which lead to synthesis of body compounds or to release energy needed for growth.

## Macro Minerals

### Calcium

Most abundant mineral present in the body, amounting to almost 40% of the total mineral mass.

Majority is present in bone, where together with phosphorous, it plays an essential part in hardening the skeleton and teeth.

In addition, calcium is a reserve of the mineral for its role in body fluids as ionic calcium, which is essential for nerve impulse transmission, muscle contraction and blood clotting.

**a. Absorption of Calcium**

Calcium salts are generally not highly soluble, which makes their absorption from the diet problematic. Several factors can enhance or inhibit absorption of Calcium.

**Factors enhancing calcium absorption**

1. Vitamin D, which causes the synthesis of a calcium-binding protein in the intestinal cells that, transports calcium into the plasma.



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2. Lactose (present in milk) also enhances calcium absorption by keeping it in a soluble form. The presence of lactose and the large amounts of calcium found in milk make this an excellent source of the mineral.
3. Other sugars and protein also enhance calcium absorption.
4. The acidic environment of the upper digestive tract also facilitates the solubility of calcium.
5. Body Need - This is the major factor governing the amount of calcium that will be absorbed during periods of growth, low calcium levels in the body fluids, and when body stores are depleted, more calcium is absorbed.
6. Other factors that enhance absorption are;
   - The acidic environment in the upper gastro intestinal tract
   - The normal gastro intestinal motility

**Factors inhibiting calcium absorption**

- Calcium absorption is reduced by phytic acid present in whole cereals, owing to the formation of insoluble calcium phytate.
- Oxalates (present in spinach, rhubarb, beetroot, chocolate, tea infusions, wheat bran, peanuts and strawberries) may also inhibit calcium absorption owing to the insoluble nature of the calcium oxalate salt
- Non-starch polysaccharides may trap some calcium making it unabsorbed in the small intestine. However, fermentation of the soluble NSP in the large intestine may release the calcium for absorption here.
- Unabsorbed fats will combine with calcium to form soaps, removing calcium from the body. This is a particular problem in steatorrhea, in which loss of vitamin D, a fat-soluble vitamin, may aggravate the problem of calcium absorption.

The skeleton is an active reservoir of calcium.

**Role of calcium in the body**



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**In bones –** Calcium is principally located in the bones. The mineral is continually laid down by osteoblasts and resorbed by osteoclasts. Bone growth happens in childhood and adolescence and maintained in adults. Changes in levels of some hormones such as cortisol or oestrogen will cause bone loss. The amount of bone declines with age. It happens earlier and rapidly in women at the time of menopause. When bone breakdown exceeds repair the bones become fragile and can easily break. This condition is known as osteoporosis.

**In blood and body fluids –** Necessary for homeostasis where levels are maintained at 2.2 – 2.6 mmol/L. When plasma calcium is low (or phosphate levels are high) parathyroid hormone is secreted. This increases calcium level by promoting synthesis of active vitamin D and thus, increasing calcium absorption from the gut, reducing calcium excretion at the kidney and stimulating calcium release from the bone. Conversely, high calcium levels cause the release of calcitonin from the thyroid gland. This inhibits bone mobilization and promotes calcium uptake into bone.

**Calcium is essential for blood clotting** where insoluble prothrombin is converted to thrombin by the action of fibrin and other clotting factors. If calcium levels are insufficient, blood will not clot.

**Muscle contraction and nerve impulse contraction** at nerve endings both involve the movement of calcium across the cell membrane, increasing intracellular levels and triggering contraction or depolarization.

**Calcium is required for the normal transmission of nerve impulses.** The calcium ions operate at the tips of nerve branches and help to excite the muscle fibre.

**Calcium activates a number of enzymes**, including pancreatic lipase (digest fat) and adenosine triphosphate (involved in the release of energy for muscular activity). Calcium also aids in the absorption and utilization of vitamin B12.

**Permeability of the cell membrane** - Calcium regulates the passage of



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substances across cell membranes

**Calcium excretion**

Calcium is lost from the body via urine and faeces, with very small amounts through sweat.

Urinary calcium represents the final adjustment of calcium levels, with the majority (up to 97%) of the calcium filtered being reabsorbed by the renal tubules.

Levels of urinary calcium are increased:

- On a high calcium intake
- By a high protein diet
- By high sodium intake
- In women at the menopause
- In old age
- By a high potassium intake
- By a high magnesium intake
- By a high phosphorous intake

Factors that help to develop peak bone mass

- Body weight – Excessively thin females with BMI below 20 may be amenorrhoeic. The absence of normal menstrual cycles and lack of oestrogen will prevent normal bone accretion. This may be a problem in girls with eating disorders. Body size is a good indicator of bone mass. Muscular development increases bone development to support movement.
- Alcohol and smoking. These both reduce bone accretion and are therefore detrimental to bone health
- Vitamin D. Adequate exposure to sunlight is important for vitamin D synthesis which is necessary for calcium absorption.
- Vitamin K is important to facilitate bone development (Osteocalcin and matrix glaprotein are dependent on Vitamin K).
- Vitamin C. This is an essential factor in the synthesis of collagen that



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forms part of the structural framework of bones.

**Maintenance of Serum Calcium Levels**

Serum calcium levels and the ratio of serum-calcium to phosphorous are held relatively constant at the expense of bones if dietary intake of calcium is not adequate. To maintain blood calcium concentration within normal limits, several hormones play an interactive role, whereby they control the amount of calcium absorbed and excreted, and also bone metabolism. The hormones include active vitamin D, PTH, calcitonin, estrogen, testosterone and other. They respond to changes in the serum calcium levels in three ways.

Active Vitamin D increases calcium absorption, mobilizes calcium from the bones and increased renal absorption to some extent

PTH also increases calcium absorption, mobilizes calcium from the bones, and increases renal re-absorption.

Calcification helps to correct high levels by preventing further bone re-absorption.

**Note:**

Altered serum calcium levels are usually due to hormonal abnormalities or may occur secondary to other clinical conditions such as inadequate secretion of PTH due to hypoparathyroidism which may be surgically induced or otherwise (e.g. in thyroidectomy, or neck dissection)

Hypercalcemia can cause nausea, vomiting anorexia, abdominal pain, constipation, polydipsia, mental changes, calcium renal stones and excessive calcification of bones and soft tissues. Coma and death may result if not treated.

Hypercalcemia may be cause by cancer excessive combined intakes of calcium and Vitamin D, hyperparathyroidism (increase secretion of parathyroid hormone) thyroid disorders and hypophosphatemia.

Re-absorptive Hypercalciuria is a condition that occurs during periods of prolonged immobility, when an excessive amount of calcium is withdrawn



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from the bones and excreted in urine, thus increasing the risk of calcium renal stone formation.

**Deficiency**

Calcium deficiency may be related to an inadequate intake or may occur secondary to mal-absorption syndromes, vitamin D deficiency or endocrine disorders. A long standing deficiency will lead to:

- Rickets and badly formed teething children
- Hypertension
- Osteoporosis is adults due to withdrawal of calcium from the bones to raise the serum calcium levels, leading (bone anthrop) and resulting in shortening of stature. The bones are also porous and brittle and fracture very easily.

**Note:** Exercises appear to reduce loss of calcium from bones; also there is clear evidence that providing the female hormone estrogen to women after menopause reduces bone loss and osteoporosis.

- Stunted growth in children
- Tetany – a condition characterized by uncontrolled muscular contractions (spasms) and failure to relax thus causing the body to stiffen with pain, due to low serum calcium levels or too much phosphorous, which causes a decrease in calcium absorption.
- Failure of blood clotting.

### 6.4.6 Food sources

All the calcium in the human body except that inherited from the mother, comes from food and water consumed. It is necessary to have adequate quantities of calcium she draws extra supplies from her bones. An entirely breastfed infant will obtain adequate calcium from breast milk as long as the volume of milk is sufficient.

A lactating mother secreting 1 litre will thus lose 300mg of calcium per day. Cow’s milk is a very rich source of calcium (richer than human milk). A litre of human milk contains 300mg of calcium than a child needs hence it is



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meant for the calf.

**Other rich sources**

Excellent Sources: milk, hard cheese, yoghurt, dark green leafy vegetables and sardines.

Good Sources: Softer cheese, ice cream and dried legumes

Fair Sources: oranges, nuts and eggs

Poor sources: most of other fruits, vegetables and grains and meats.

**Calcium Content of Various Milks**

| Sources of Milk | Calcium Content (mg/10ml) |
|---|---|
| Human | 32 |
| Cow | 119 |
| Camel | 120 |
| Goat | 134 |
| Water Buffalo | 169 |
| Sheep | 193 |

The calcium content of drinking water varies from place to place. Hard water usually contains high levels of calcium.

**Recommended dietary requirements**

It is not easy to state categorically the human requirements for calcium, because there are several factors influencing absorption and considerable variations in calcium losses among individuals. Need for calcium is increase during pregnancy and children require more calcium because of growth. Those on high protein diets require more calcium in the diet. The following are recommended levels of daily calcium intake:

| | |
|---|---|
| Children ages 1 to 10 years | 800mgs/day |
| For men and women aged 18 to 24 years | 1200mgs/day |
| Adults aged 25 years and above | 800mgs/day |
| Pregnant and lactating mothers’ | 1200mgs/day |

**Phosphorus**

After calcium, the most abundant mineral in the body is phosphorus. It



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constitutes 22% of the minerals in the adult human body or 1% of the body weight. It is a major constituent of bones and teeth or 85-90% of total body's phosphorus. The remainder is distributed in body tissues with half in muscles. It is part of the nucleic acids; DNA and RNA which are found in the nucleus of every living cell. As a phospholipid, it is a structural component of the cell wall and participates in all biological reactions.

**Functions of phosphorus**

1. Regulation of energy release from oxidation of carbohydrates, fats and proteins: Energy is stored as ATP (adenosine triphosphate) and when energy in needed, ATP is changed to ADP (adenosine diphosphate).
2. Absorption and transport of nutrients: Phosphorus in the form of phosphate makes it possible for nutrients to cross cell membranes or be carried in the blood. This process is known as phosphorylation and occurs in several areas: absorption in the intestines, release of nutrients from the blood stream to the intercellular fluids, uptake into the cells, and uptake by organelles of the cell. When glycogen is released from the liver or muscles for energy, it appears in the blood combined with phosphorus.
3. Part of essential body compounds: Some vitamins and enzymes can function only when they are phosphorilated e.g. thiamin pyrophosphate.
4. Essential part of DNA and RNA which are essential for cell reproduction.
5. Calcification of bones and teeth.
6. Regulation of acids-base balance: Serves as buffers since they can take additional hydrogen ions.

**Absorption and metabolism of phosphorus**

It is released from food by enzymes known as phosphatases. It is absorbed with the help of vitamin D. The level in blood is regulated by the parathyroid gland.

**Dietary sources of phosphorus**

Phosphorus is present in most foods especially those rich in protein e.g.



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meat, poultry and fish, eggs dairy products and cereal products.

**Deficiency of phosphorus**

Phosphorus deficiency is unknown in humans but may affect people who consume large amounts of antacids especially those with aluminium hydroxide which interfere with phosphorus absorption. Premature infants may need more phosphorus than what is provided by breast milk.

**Magnesium**

The concentration of magnesium in the body is roughly 30mg; 60% of which is found in the bone. About a third of this is closely bound to phosphate and the remainder is on the surface of the bone from which it is readily mobilized to maintain normal blood and tissue levels. The blood contains about 1% of total body magnesium, a third of this is bound to the protein albumin, a third is free and a third is part of other compounds. It is absorbed in the small intestines with the help of a carrier. About half of the dietary intake is absorbed. Absorption is reduced by presence of calcium and alcohol phosphate, phytates and fats. Absorption is increased by dietary vitamin D and lactose. Excretion is regulated by the kidneys and reabsorption is facilitated.

**Functions of magnesium**

a. Within the cell, magnesium acts as a catalyst to many biological reactions especially those taking place in the mitochondrion of the cell. Magnesium is involved in cell reactions involving expenditure of energy from carbohydrate, fat and protein; and nucleic acid metabolism in those reactions related to the synthesis, degradation and stability of the genetic material including DNA.
b. It is necessary for conduction of nerve impulses and for normal muscular contraction. In these roles, magnesium and calcium play antagonistic roles. Calcium stimulates whereas magnesium relaxes. A very high intake of magnesium could lead to coma and heart failure.
c. Magnesium is also involved in bone formation together with phosphorus



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and calcium.

**Dietary sources of magnesium**

Dark green vegetables are the best sources of magnesium followed by legumes, sea food, nuts, cereals and dairy products. Water contains good amounts especially soft water; sea water has very high levels.

**Deficiency of magnesium**

Magnesium deficiency is associated with starvation, persistent vomiting and trauma of surgery or diarrhoea. A deficiency leads to weakness, confusion, growth failure in children. Severe deficiency leads to tetany (uncontrolled neuromuscular tremors which progress to convulsive seizures). Deficiency also leads to vasodilation (increase in diameter of blood vessels) and skin changes.

**Sulphur**

Sulphur represents only 0.25% of total body weight. It is concentrated in cytoplasm of the cell. High concentrations are found in the hair, skin and nails, hence the smell of Sulphur dioxide when they are burned. The Sulphur containing amino acids; cysteine and methionine are characteristic of the protein found in the tissues. Sulphur gets to the body through these Sulphur-containing amino acids.

**Functions of sulphur**

1. Sulphur in combination with hydrogen plays an important role in metabolism.
2. Involved in formation of blood clots.
3. Involved in the transfer of energy.
4. It is part of at least 3 vitamins; thiamin, pantothenic acid and biotin. These are part of coenzymes which activate several enzymes.
5. Compounds containing Sulphur act as detoxifying substances by combining toxic substances and converting them into harmless compounds that are excreted by the kidney.
6. Sulphur is also necessary for collagen synthesis and the formation of



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many mucopolyssacharides.

**Fluorine**

Fluorine is found mainly in the teeth and skeleton. Traces of fluorine in the teeth help to protect against decay. When consumed in early adulthood, it becomes part of the enamel, making it more resistant to the organic acids formed from foods that get stuck in between the teeth. Fluorine is also known to harden and strengthen the bone especially later in life and thus slow or inhibit the development of osteoporosis.

**Mode of action**

Fluorine forms crystals of flourapatite which replaces crystals of hydroxyapatite normally deposited during tooth formation. Flourapatite in the tooth enamel is more resistant to cavity forming acids. Absorption is known to occur in the mouth through the tooth surface. Intakes of 8ppm in adult life protects against osteoporosis (a condition where there is a reduction of the bone density due to erosion of calcium).

**Dietary sources of fluorine**

- The main source is water. If it contains 1ppm (part per million), this will be adequate.
- Other sources include small fish when consumed whole.
- Tea has a high content.
- Foods grown in high fluoride water will have more than fresh foods.

**Toxicity**

Too much fluorine intake causes browning of teeth enamel and excessive amounts may cause dental fluorosis in which the teeth become mottled.

**Microelements**

They are required in small amount by the body compared to the macro elements.

**Iron**

Iron is found in every living cell. The total body content is about 5grams;



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varying slightly with age, sex, body size, nutritional status and general health. It is the only nutrient for which adult women have greater requirements than adult men.

**Absorption and distribution of iron**

The body uses several mechanisms to regulate iron absorption because the body cannot easily eliminate excess iron once it is absorbed.5-10% of iron in food is absorbed in healthy people.10-20% of iron in food is absorbed in by people with iron deficiency. The extent of iron absorption depends on a variety of factors, the most important being the amount of current iron stores in the body. Iron in food occurs in several forms which differ in their absorption by the body. Iron that is part of hemoglobin and myoglobin molecules in animal flesh is about 40% of total iron present called **heme iron** and is readily absorbed more than twice as efficiently as simple elemental iron known as non-heme iron**. Non-heme** iron is provided from plant sources and elemental iron components of animal tissues. It is less efficiently absorbed than heme iron. The difference in absorption of heme and non heme iron make animal flesh a rich source of dietary iron, considering both its iron content and the increased efficiency of absorption of the heme iron present. Consuming heme iron and non heme iron together increases non-heme iron absorption e.g. eating meat with vegetables and grain products.

**Factors affecting iron absorption**

These include the following;

**a. Body needs**

A normal person with normal hemoglobin levels absorbs only 2-10% of dietary non-heme iron and 23% of heme iron. Iron deficient individuals may absorb as much as a third of dietary iron from heme and a fifth from non-heme. Need for iron is increased with exercise because the production of red blood cells increases.



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**Form of iron**

Ferrous iron is absorbed better than the ferric iron. Organic acids such as vitamin C moderately increase non-heme absorption by adding an electron to ferric iron ($Fe^{3+}$) yielding ferrous iron ($Fe^{2+}$). Vitamin C then forms a complex called chelate with $Fe^{2+}$ thereby enhancing absorption. As such iron supplements should be taken together with vitamin C rich drinks such as fresh orange juice.

**b. Composition of meal**

When meat is eaten with a vegetable and grain products, the absorption of non heme iron present is improved. A protein factor in meat, fish and poultry (MFP factor) facilitate non heme absorption. This factor appears to contain amino acids such as cysteine that bind iron to enhance absorption.

c. **Acid in the stomach**

Acid also plays an important role in iron absorption by promoting the conversion of $Fe^{3+}$ to $Fe^{2+}$ and by solubilizing non heme iron. The decreased production of stomach acids experienced by many elderly people can lower their iron absorption and ultimately their body stores of iron.

**Dietary factors**

Several dietary factors **interfere** with the body's ability to absorb iron and these include;

**Phytic acid and oxalic acid** in vegetables which bind iron thus reducing its absorption.

**Dietary fibre intakes above 35g/day** bind iron and other trace elements.

**Polyphenols such as tannins** found in tea and related substances found in coffee also reduce iron absorption. People trying to rebuild iron stores are advised to reduce coffee and tea consumption, particularly meal times.

**Calcium supplements** in amounts greater than 300 mg/d can reduce iron absorption. Calcium supplements should be taken several hours before or after an iron rich meal.



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**Function of iron**

1. The major role of iron is to permit the **transfer of oxygen and carbon dioxide** from one tissue to another. Iron does this as part of haemoglobin in the blood and myoglobin in the tissues.
2. **Blood formation**: Haemoglobin is an essential component of the red blood cells which are formed in the bone marrow when stimulated by erythropoietin, a hormone synthesized by the kidney. Erythropoietin is released in response to a decrease in oxygen occlusion in the blood, blood loss or binding of carbon dioxide to red blood cells. It targets the bone marrow to increase production of red blood cells.
3. Other functions of iron: A small amounts of iron are incorporated into tissue enzymes. In a dietary deficiency, the level of these enzymes drops before the haemoglobin level drops.

Iron is also important in many other functions such as;

- Catalyzing the conversion of beta carotene into vitamin A.
- The synthesis of purines which form integral part of nucleic acid
- The removal of lipids from the blood.
- Collagen synthesis.
- Antibody production needed for proper immune formation.
- Detoxification of drugs in the liver.

**Iron needs**

Iron needs differ for different age groups and sex. Pregnant women and adolescent girls need twice as much as iron compared to adult men. Iron needs are determined by need to replace losses through sweat, hair, nails, dead skin cells, mensturation or other forms of bleeding. Growth increases iron needs due to increased blood volume and tissue mass. The adult recommendations for iron are 10mg/d for men and 15mg/d for women to account for menstruation losses.

Pregnant women require 30mg/d daily. This takes care of the increased placenta blood volume and the foetal needs. Newborns are born with



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enough iron stores to last for the first 4-6months until supplementary feeding is introduced. Human milk is a poor source of iron 15mg/d. Infants and children: Reserve of iron last 4–6months after delivery hence complementary foods introduced at 6 months need to be rich in iron.

**Dietary sources of iron**

Liver is the richest source, lean meat, chicken, eggs, lean pork, dark green vegetables, legumes, nuts, molasses and cereals.

**Iron deficiency**

Continued iron deficiency leads to nutritional anemia caused by lack of dietary essentials involved in haemoglobin formation or poor absorption of these dietary essentials. Some types of anaemia have been reported to be caused by a lack of either dietary iron or high quality protein; lack of pyridoxine (vitamin $B_6$), which catalyzes the heme portion of the haemoglobin molecule, by a lack of vitamin C which influences the rate of iron absorption and the release of iron from transferring to the tissues; and by lack of vitamin E, which affects the stability of the red blood cell membrane. Copper is not part of the haemoglobin but aids in its synthesis by influencing the absorption of iron, its release from the liver, or its incorporation into haemoglobin.

Nutritional anaemia is characterized as both **hypochromic** and **microcytic**. Hypochromic suggests a lack of red pigment haemoglobin and microcytic indicates the presence of small red blood cells. When red blood cells synthesis is reduced, the amount of oxygen carried to the blood is decreased. There are many types of anaemia but the major type found in the world is iron deficiency anaemia. It is common in adolescent girls (growth and menstrual demands), young infants and women of child bearing ages.

**Iron toxicity**

Excess intake may lead to hemosiderosis i.e. increased iron reserves due to excessive supplements. Hemochromatosis: A genetic disorder of iron



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metabolism characterized by increased absorption, saturation and deposition of iron in the liver tissues.

**Iodine**

Iodine is present in the body in minute amounts, 15-23 mg and 75% of this is concentrated in the thyroid gland that uses it to synthesize the hormone, thyroxin. The remainder is in other tissues particularly in the salivary, mammary and gastric glands and in the kidney. It is present in food as iodide and other non-elemental forms. Iodine was linked to the presence of an enlarged thyroid gland (goiter) during World War I.

**Functions of iodine**

1. Regulation of growth and development as part of thyroxin hormone. Thyroxin can stimulate metabolism by as much as 30%.
2. Conversion of carotene to vitamin A.
3. Synthesis of protein.
4. When thyroxin levels are normal, absorption of carbohydrates from the intestines is more efficient.
5. Synthesis of cholesterol.

**Recommended daily allowances:**

For adults 150mg/day

**Dietary sources of iodine**

These include: Salt water, fish, sea foods, iodized salt, molasses, dark green leafy vegetables especially spinach leaves.

**Deficiency of iodine**

**a. Goitre**

This is a condition characterized by swelling in the neck as a result of the enlargement of the thyroid gland. The gland enlarges as it attempts to compensate for lack of iodine essential to its major role in the synthesis of thyroxin. The pituitary gland hormone that stimulates the thyroid hormone production, thyroid stimulating hormone also stimulates the growth of the thyroid gland. Too much growth of the gland causes difficulties in breathing.



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Goitre is also associated with the consumption of goitrogens (substance in foods such as the cabbage family and water that interfere with thyroid gland metabolism and thus may cause goitre if consumed in large amounts). Defects in the enzymes necessary for the synthesis and release of thyroxin can also lead to goitre. Goitre is common in areas where the soil is poor in iodine due to leaching. Goitre incidences are six times more in females than males with adolescent girls and pregnant women being more susceptible.

**b. Cretinism**

If an expectant mother consumes iodine deficient diets during the early pregnancy, her infant may be born with a short stature (dwarfed) and with mental retardation. Maternal iodine needs take precedence over foetal needs. The retarded body growth is referred to as cretinism characterized by physical drawfness, mental retardation, thick-dry skin and enlarged protruding stomach.

**c. Myxedema**

This is a condition seen in adults who have had symptoms of hypothyroidism throughout their development and growth period. It is characterized by coarse sparse hair, dry, yellowish hair, poor tolerance to cold and low husky voice.

**d. Hyperthyroidism**

This is where basal metabolic rate is accelerated by as much as 100% above normal and is associated with over activity of the thyroid gland and experience nervousness, weight loss, increased appetite, intolerance to heat, tremors and protruding eyeballs.

## Zinc

Zinc is present in all body tissues and fluids. The total human body zinc content has been estimates to be 30 mmol (2grams). Skeleton muscle accounts for approximately 60% of the total body content and bone mass, with a zinc concentration of 1.5-3.0mol/g (100-200g/g) for approximately 30%. The concentration of zinc in lean body mass is approximately 0.46 mol/g (30(g/1). Plasma zinc has a rapid turnover and represent only about 0.1% of total body zinc content. This level appears to be under close homeostatic control. High concentrations of zinc are found in the choroids of the eye (4.2 mol/g or 274 g/g) and in prostastic fluids (4.6-7.7 mmol/l or 300-500mg/l). Zinc is lost through



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faeces, urine and shed tissues. High tissue losses are from the skin, the mucosal cells, menstrual fluids and semen.

**Absorption of zinc**

Like iron absorption, zinc absorption is influenced by the types of food ingested. More is absorbed when animal protein is consumed. **Fibre and phytates inhibit zinc absorption**.

**Functions of zinc**

Over 300 enzymes require zinc as a co-factor for optimal activity. Adequate zinc is necessary to support many bodily functions such as;

a. It is involved in nucleic acid synthesis and function (some factors that control expression of genes contains zinc-rich regions that bind DNA).
b. It is involved in protein metabolism, growth and wound healing.
c. Immune function.
d. Development of sexual organs and bone.
e. Involved in storage, release and function of insulin.
f. Cell membrane structure and function.

**Recommended daily requirements for zinc**

The adult recommended daily requirement is 15 mg/day for men and 12 mg/day for women.

**Dietary sources of zinc**

In general, protein-rich diets are also rich in zinc. Lean meats especially beef and other red meats, eggs and shell fish are the best sources. Cereals and legumes contain significant amounts of zinc but contain phytic acid and other substances that can interfere with intestinal absorption of zinc.

**Zinc deficiency**

Zinc deficiency is associated with;

- Severe growth retardation and arrested sexual maturation.
- Altered taste acuity.
- Impaired immune function.
- Slow wound healing.
- Impaired central nervous system function.

## Selenium



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Selenium is deposited in all body tissues except fat. The highest concentrations occur in the liver, kidney, heart and spleen. Serum levels are about 0.22 µg/dl. Most selenium in foods is bound to derivatives of the amino acids; methionine and cysteine. Because these substances are readily absorbed, bioavailability of selenium is considerably higher than that of zinc and iron. About 50%-100 % of dietary selenium intake is absorbed. In addition, since no physiological mechanism appears to control selenium absorption, selenium has a definite potential for toxicity. Most selenium is excreted via the urine and faeces.

**Functions of selenium**

a. It functions as an integral component of an anti-oxidant enzyme; glutathione peroxidase that protects cells and lipid membranes against oxidative damage.
b. It also acts as a structural component, incorporated into the protein matrix of the teeth.

**Recommended daily allowances for selenium**

The recommended daily allowance estimated to be safe and adequate for an adult is about 50-200 µg/day.

**Dietary sources of selenium**

Food sources vary with the selenium soil content. Good sources include sea food, legumes, whole grains, low fat meats and dairy products with smaller amounts in vegetables.

**Selenium deficiency**

The signs and symptoms of selenium deficiency include pain, muscle wasting and **cardiomyopathy**, a form of heart disease.

**Selenium toxicity**

High doses (a milligram or above daily is toxic. Selenium toxicity causes vomiting, diarrhoea, loss of hair and nervous system.

## VITAMINS

Are a type of micronutrients that are required by the body in small amounts. They are organic compounds made of carbon, hydrogen, oxygen and sometimes nitrogen or other elements that are essential for normal functioning, growth, and maintenance of the body. They are nutrients that are essential to life.

Vitamins are individual molecules that do not provide energy but are needed for the metabolism of fuel molecules (carbohydrates, proteins and lipids) for energy.



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They are categorized based on their solubility into;

A. Fat soluble vitamin
B. Water soluble vitamins

**A. Fat-Soluble Vitamins**

These are soluble in fats**.** They are absorbed by the intestinal cells with dietary fat in chylomicrons that enter the lymphatic system before circulating in the blood stream.

They include vitamin A, D, E and K. They are stored in the liver and adipose tissues from where they can be drawn upon in times of need. They can be toxic when consumed in large doses over a prolonged period especially in the case of vitamin A and D. They do not have to be taken every day because the body can retrieve them from storage as needed.

They are found in fat and oil portion of food e.g. vitamin E in the germ of cereal grains.

These vitamins include;

**i) Vitamin A**

This vitamin is held in the body in three active forms that are collectively known as **retinoids** (preformed vitamin A). These forms are:

- **retinol** (alcohol form)
- **retinal** (aldehyde form- organic compound containing CHO formed from oxidation of alcohol)
- **retinoic acid** (acid form).
- Vitamin A also exist as a provitamin A form known as **carotenoid** which is present in varying types such as **beta(β) carotene, alpha (α) carotene, lutein(deep yellow)** and **lycopene(red carotenoid found in tomatoes and red berries)**
- **Retinol** and **retinal** are freely **interconvertible** in the body. The body can convert ***retinal*** *to* ***retinoic acid irreversibly***.
- **Retinol** is the most important compound in the vitamin A family. The body can convert **α –carotene** and **β-carotene** to **retinols** thus are important provitamin A caretonoids. **Lycopene** and **lutein** have **no vitamin A** activity thus are referred to as **n*on-provitamin A carotenoids.***

These carotenoids act as **antioxidants** since they interfere with the damaging effects of free radicals and prevent carcinogens from causing cancer in the body.



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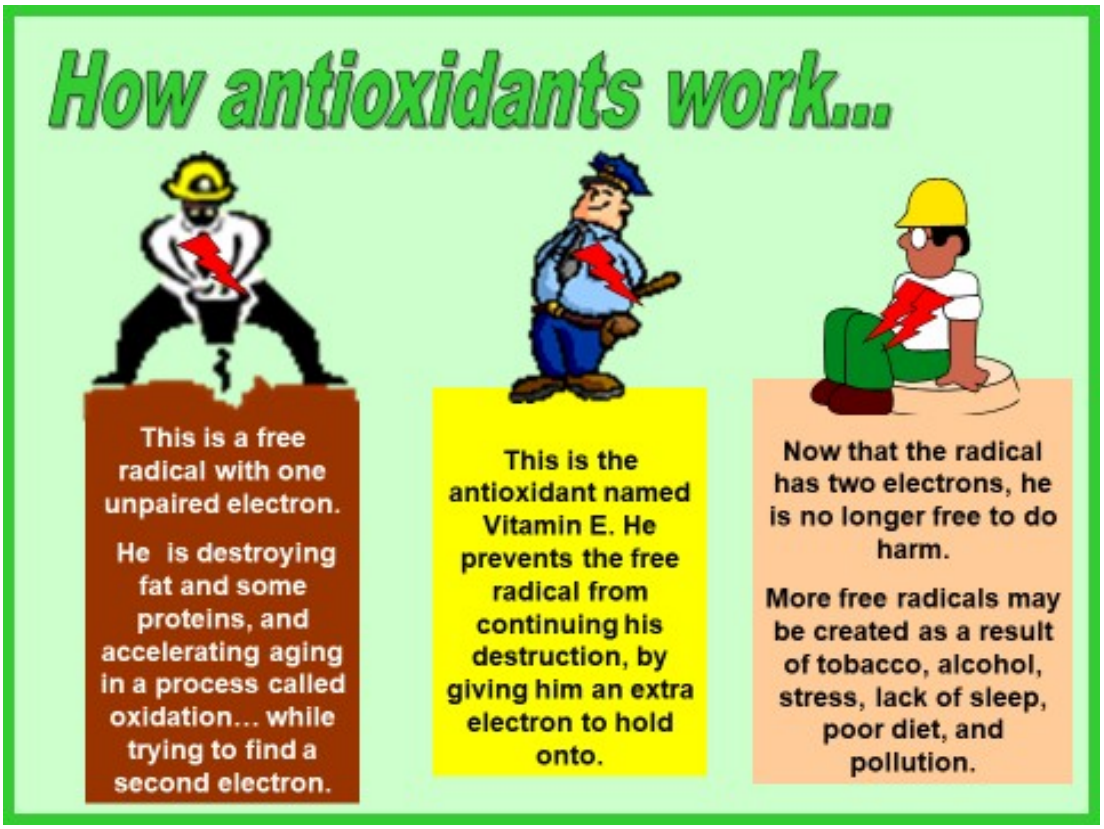


Vitamin A is stored in the body as **retinyl esters (i.e palmitic acid esters of retinol**).
The liver in a well nourished person can store more than 90 percent of vitamin A present in the body.
The liver releases retinol when required. When released it is carried by retinol –binding protein in blood to different organs where it is required. Within the target cell, retinol is converted to retinal or retinoic acid as needed.

**Functions of Vitamin A**

**i.** **It is essential for night and colour vision** in that it is activated to **retinal** which helps in the formation of ***rhodopsin pigments*** in the **retina** of the **eye** that helps in the **transmission of visual impulses to the brain**. This thus helps the eye to adapt to dim light. During colour vision, rhodopsin in rod cells and iodosin in cones are formed which undergo transformation cycle.

**NB** People with deficiency of vitamin A experiences the inability of the eyes to adjust to dim light or to gain vision quickly after exposure to a flash of bright light a condition known as **night blindness (**nyctalopia**).** This condition is due lack of vitamin A as a result of *which regeneration of rhodopsin slows down.* Lack of vitamin A affects rod cells before it affects cone cells so as vitamin A deficiency worsens night blindness emerges before colour blindness



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ii. **Cell differentiation** (This is where stem cells develop into high specific types with unique functions)- **Retinoic acids** plays important role in normal cell differentiation. It interacts with the receptors in the cell nucleus as a result of which stem cells e.g. epithelial cell differentiate to mucous secreting goblet cells.

iii. **Immune Functions-** All forms of vitamin A influence *the immune system by maintaining the cell of the epithelial tissue which is the* ***first line of defence*** against microbial attacks. It also supports the T-Lymphocytes which maintain the ability of the body to mount immune response against invading microorganism

iv. **Reproduction-** Retinal and retinol support the production of sperm in men. These forms of vitamin A also help in maintaining fertility in women by supporting the production of reproductive tract secretions.

**v.** **Bone Health-**Vitamin A is required in bone remodeling process and in the proper development of immature bone cells.

**Deficiency of vitamin A**

**N/B-**Vitamin A deficiency results from general protein energy malnutrition in infants and young children.

1. Early symptom of vitamin A deficiency is **night-blindness** (failure to adjust to dim light or to recover after flash of bright light).
2. As it worsens there is lack of normal differentiation of epithelial cells and reduced formation of mucus secreting cells; as mucus production drops, the cornea, conjunctiva and mucus membrane lining the inner surface of the eyelid become dry a condition known as **xerosis and ulcers.**
3. As this condition deteriorates there is the formation of white irregular patches known as **Bitots spots** which later develop into ulcers i.e. corneal ulcers. If the disease is not treated corneal ulcers can lead to "melting" or "wasting" of the cornea, a condition known as **keratomalacia**. It is a non-reversible condition. This condition can result in **corneal scars.** These symptoms associated with vitamin A Deficiency develops towards total blindness if not treated and are collectively known as **xerophthalmia.**



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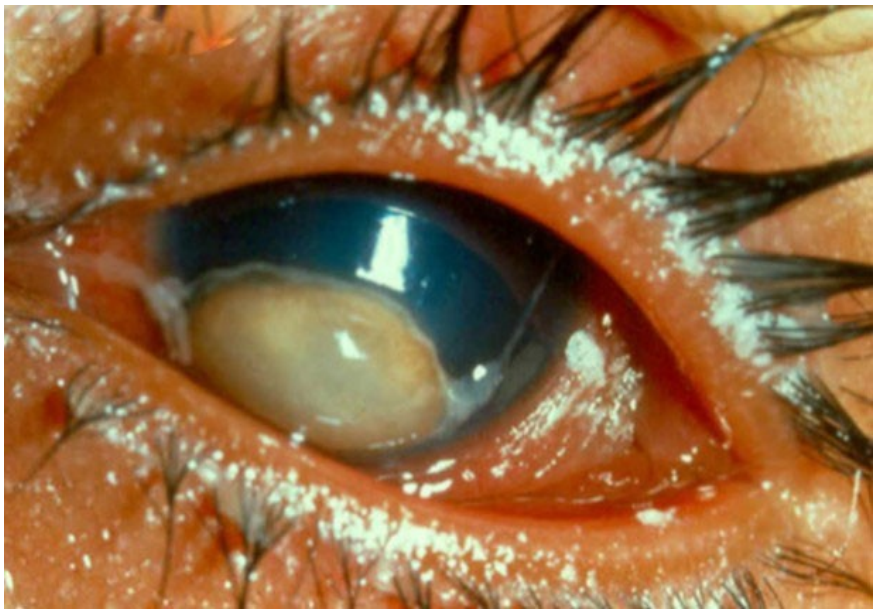

4. **Follicullar Hyperkeratosis**

This occurs due to lack of retinoic acid. This is a condition where hair follicles on the skin become plugged with keratin as a result of which the skin become rough and bumpy which cause thickening of the palms and soles as well the elbows, knees, wrists and ankles (i.e. flexus areas of the skin )

This condition also affects other types of epithelial cell suppressing their ability to secrete mucus; it particularly affects the mouth, respiratory tract, urinary tract, female genital tract, seminal vesicles of the testicle and glands of the eye making them vulnerable to infection

5. **Vitamin A deficiency** also affects production of sperms in men. Women become infertile possibly due to disruption in the production of reproductive tract secretion (retinal and retinol)

6. **Hyperkeratinization** near sensory receptors causes a loss of sense of taste and smell
7. **Reduced immune function;** Lack of mucus secretions due to deficiency of vitamin A



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makes a person vulnerable to Bacterial, Parasitic or Viral Infection. Vitamin A deficiency also reduces the number of T- lymphocytes. Children with lack of vitamin A have a high risk of diarrhea, respiration tract infection and measles.

8. It retards growth and development and leads to bone deformities.

**Vitamin A Toxicity**

Excess vitamin A consumption can lead to toxicities characteristic of the following symptoms;

i. Headache
ii. Vomiting
iii. Double vision
iv. Hair loss
v. Bone abnormalities
vi. Liver damage
vii. Birth defects during pregnancy
viii. Skin disorders
ix. Excess vitamin A prior to conception and in the first trimesters can cause spontaneous abortions.

**Recommended Daily Allowance of vitamin A**

- Adults retinol- is 600µg while β-carotene is 2400µg
- Lactating women require retinol-950µg while β-carotene is 3800µg
- Infants retinol 350µg while β-carotene is 1400µg



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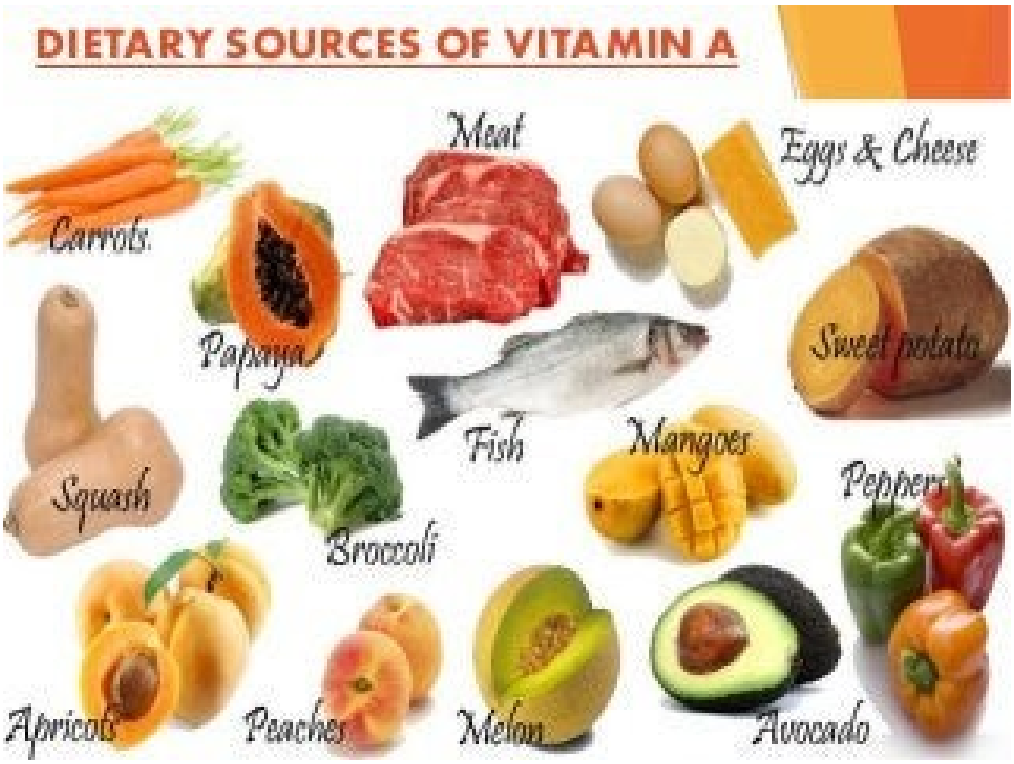


**2. VITAMIN D;**

Vitamin D is a unique vitamin that under optimal conditions can be synthesized by the body in adequate amounts. Sunlight is capable of converting the precursor(component from which another substance is formed) of vitamin D on the skin into its active form. The precursor on the skin is called **7 – dehydrocholestrol**. The precursor in the **liver** is converted to **25–hydroxy vitamin D** and in the **kidneys** to **1, 25 – dihydroxy vitamin D.**

**N/B** People with exposed skin e.g. hands, face and legs are advantaged in that they only need between 5 – 10 minutes of exposure and they meet the requirements of vitamin D per day. Few people are unable to meet their daily vitamin D requirements therefore the need for a dietary source.

There are few foods that provide vitamin D naturally e.g. liver, fatty fish and egg yolk. There are foods that are fortified with vitamin D e.g. all milks, most ready to eat cereals e.g. weetabix and some brands of orange juice, yoghurt, margarine e.t.c.



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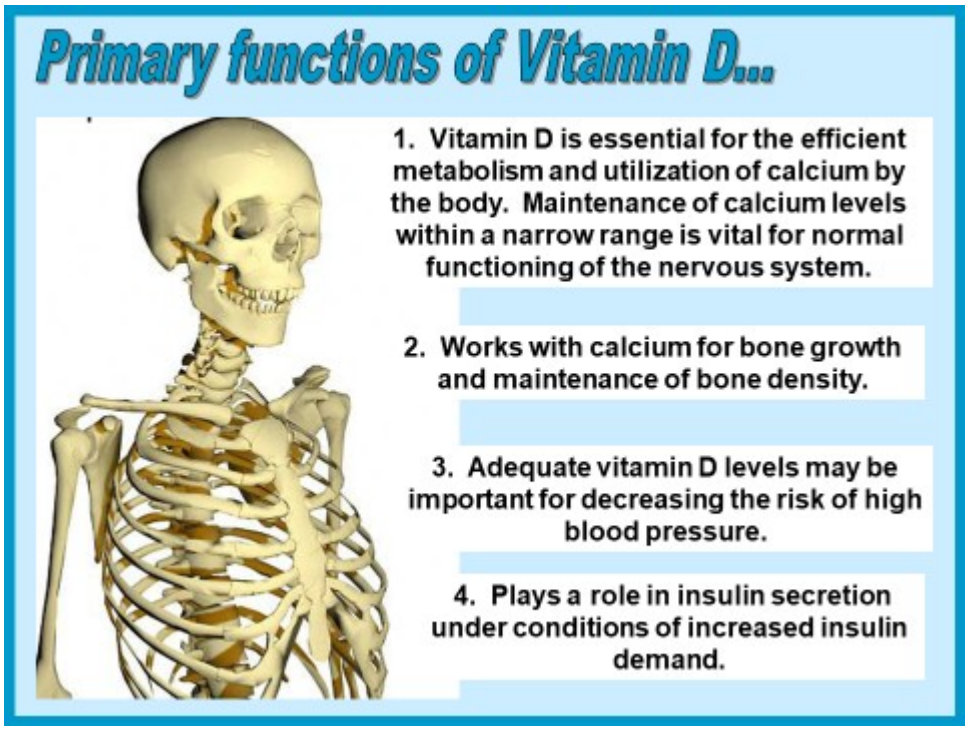


**Deficiency of Vitamin D**

A. In children it results in **rickets** with symptoms such as ;

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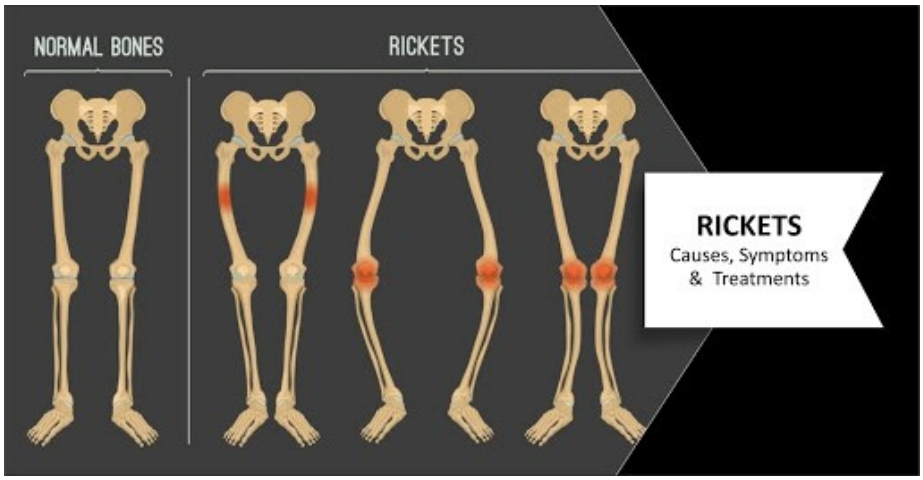


- Enlarged wrists and ankles
- Poor structural developement which affects long bones.
- The ends of bones become soft and bulky as characterised by bow-legs and knocked knees.
- There is extreme stunted growth
- Deformities in the bones of the skull, chest and spine.
- Teeth may erupt late and may be deformed
- There is a decrease in serum phosphorous and an increase in serum alkaline phosphates and excretion of inorganic phosphorous in urine.

B. In adults it results in a condition called **oesteomalacia** especially in female during pregnancy and lactation and in those protected from sun light e.g. Muslims. The pelvic bone show characteristic deformities.



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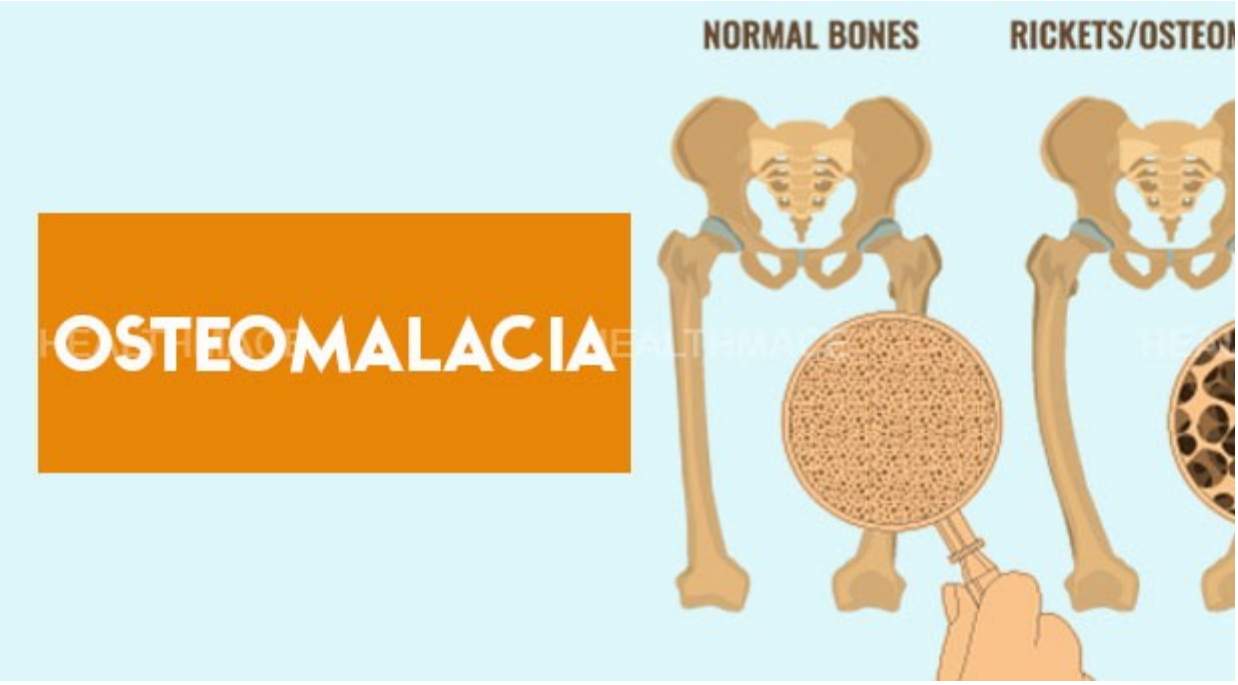
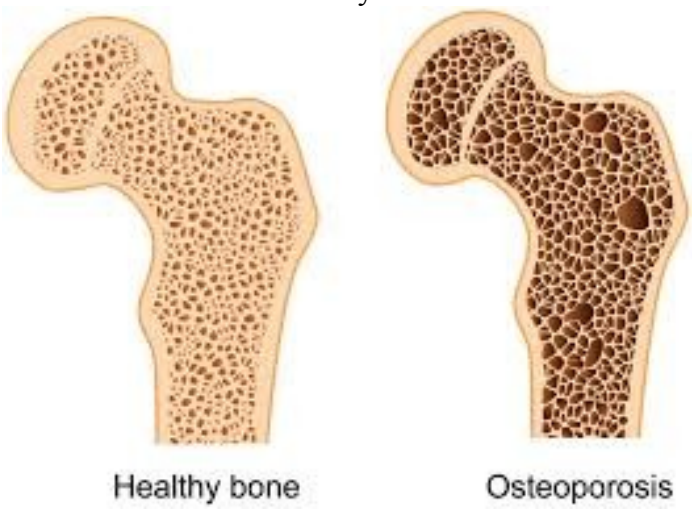



Vitamin D deficiency is also associated with **osteoporosis** which is increased bone turn over or reduced bone density.




**Toxicities associated with vitamin D**

Since the normal requirements of vitamin D are 10 – 25 micrograms per day; any intake exceeding 50 micrograms may lead to toxicities which manifests themselves through excessive calcium absorption which may lead to the following symptoms;

- Hypercalcemia
- Kidney stones,
- Kidney calcification,
- Kidney failure,
- Soft tissue calcification for example lungs, heart and blood vessel calcification that can



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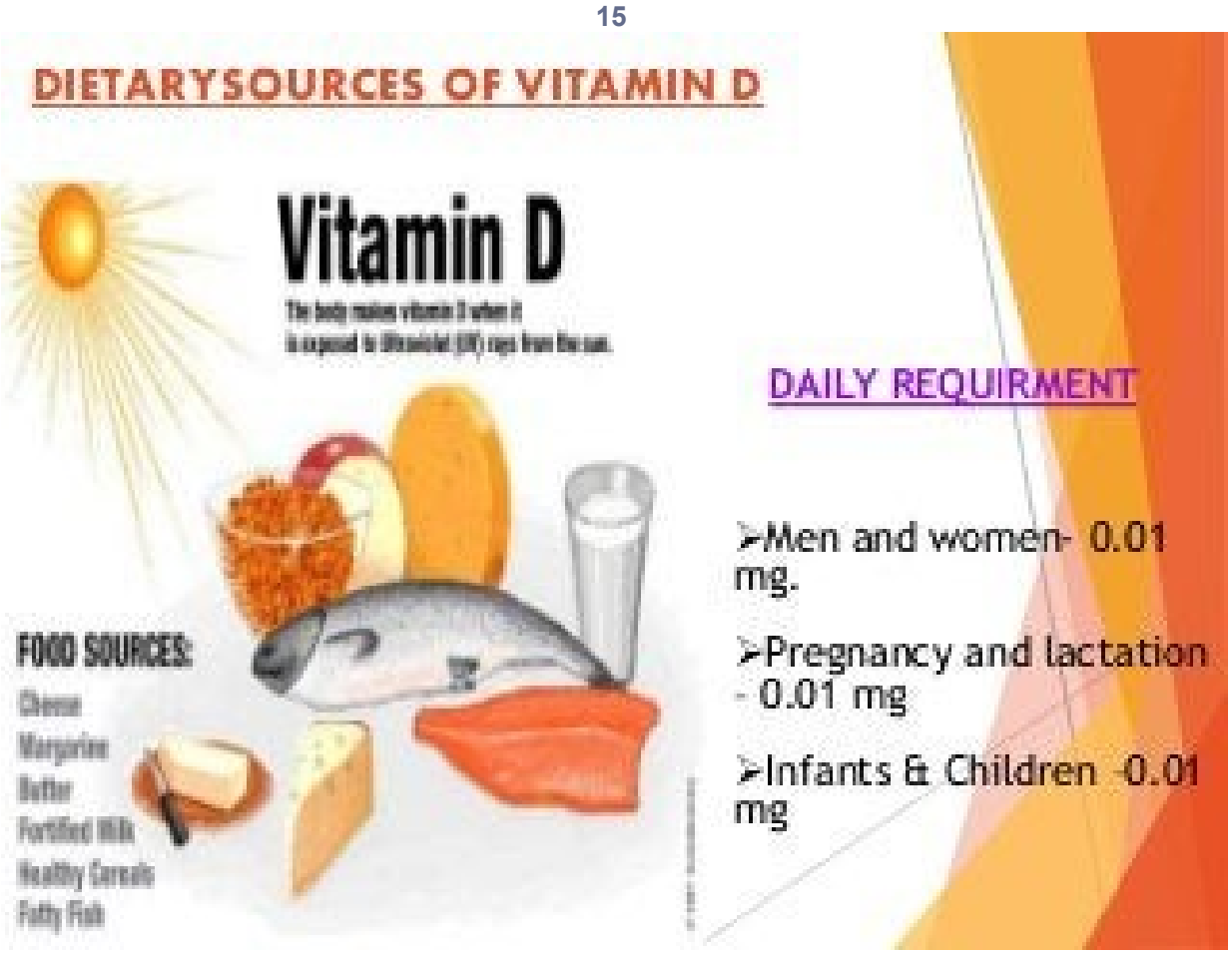
be fatal.

- Prolonged hypercalcemia also affects the nervous system causing depression as well GIT disturbances e.g. nausea, vomiting and loss of appetite.




**3. Vitamin E**

Vitamin E is a generic term that describes a group of **at least 8 natural occurring compounds**. It has **phytol** and **methylhydroquinone.** It has a set of two compounds namely **tocopherols and tocotrienol**. **Tocopherol** is of four types **α, β, γ and δ.**

**α tocopherol** is the most common active form of vitamin E available in food. Vitamin E is absorbed together with fats in the small intestines and incorporated into chylomicrons and enters blood stream via the lymphatic system. Unabsorbed vitamin E is excreted in faeces. Vitamin E is **not stored in the liver**. The Adipose tissues store 90% of vitamin E in the body and the rest is found in the cell membrane in every tissue where it functions as an antioxidant and protects cell membranes in all parts of the body from damaging effects of free radicals. Its activity is enhanced by other antioxidants e.g. Vitamin C and Selenium.

**Functions of vitamin E**



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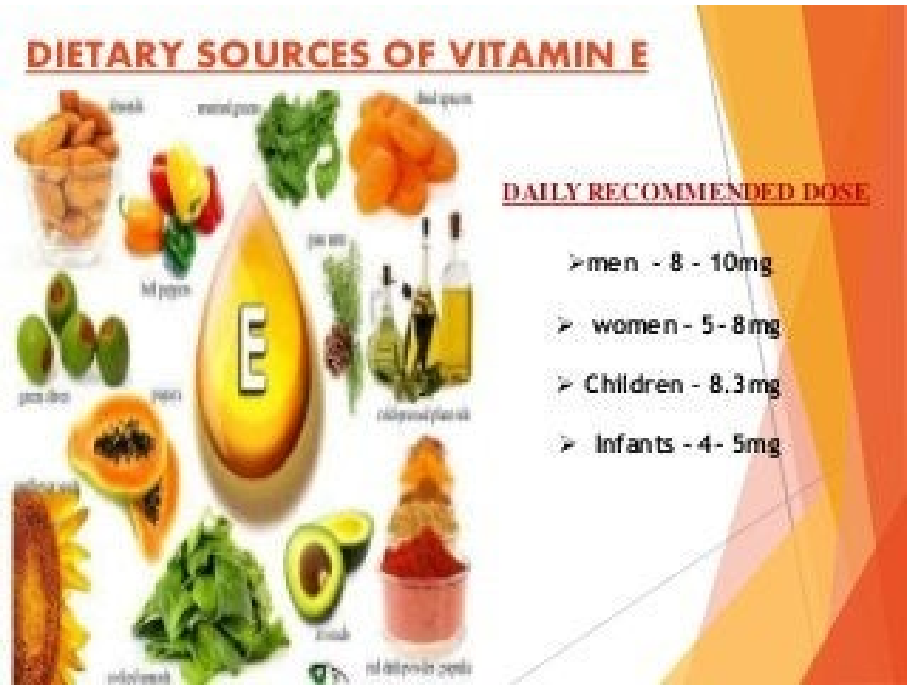
1. It acts as an **antioxidant** in the body where it protects polyunsaturated fatty acids (PUFA) and other lipid molecules from oxidative damage and defends cells against the attack of free radicals. It
2. It is required for cellular respiration through the **electron transport chain**
3. It increases the synthesis of heme by enhancing the activity of enzyme δ-amino-levulinic acid (ALA) synthase and ALA dehydratase.
4. It is required in the proper synthesis of nucleic acids
5. It delays the onset of cataracts
6. Its required for the proper storage of creatine in the skeletal muscles
7. It is required for the absorption of amino acids in the intestines.




**Vitamin E deficiency**

Vitamin deficiency is a rare condition that is experienced by people with fat malabsorption syndrome e.g. cystic fibrosis.

- It leads to **premature hemolysis**( rapture of the RBCs)
- **Retinopathy;** defective functioning of the retina of the eye which can cause permanent blindness in infants.

**Vitamin E toxicity**

Large doses of vitamin E can be hazardous to people taking anti coagulant medication. E.g. warfarin and aspirin.



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**4. Vitamin K**

Vitamin K was derived from a Danish word **Koagulation**. It was discovered in 1934 by a Danish scientist named Dam. It is a nutrient that plays a role in **blood clotting.** It refers to a family of compounds known as **naphthoquinones**.

**Forms of vitamin K;**

1) **Phylloquinone/phytylmenaquinone (vitamin $K_1$)** from plant sources.
2) **Menaquinones** (K 2) from animal sources and synthesized in the intestines by bacteria.
3) **Menadione** (vitamin K 3) is a synthetic form of vitamin K that is a water soluble compound.

**N/B-Phylloquinone** is the most active form of vitamin K.

Approximately half of vitamin K requirements are supplied by the diet; the other half is by bacterial synthesis in the G.I.T.

**Functions of Vitamin K**

1) It plays a role in the formation of a blood clot through the formation of proteins e.g. Prothrombin factor II, Proconvertin factor III, Plasma thromboplastin component factor IX and Stuart factor X.

   **Vitamin K** converts **preprothrombin** to **prothrombin** in the **liver** by **binding carbon dioxide** to **glutamic acid** in the protein to a new amino acid called **Gamma-carboxyglutamate acid.** Once formed it provides **calcium binding capacity** to the **protein** which allows **prothrombin** to be changed to **thrombin** which catalyze the conversion of **fibrinogen**, a solube protein in blood to **fibrin**.
2) Vitamin K also helps in bone formation. It helps in the **carboxylation of osteocalcin** (a bone **protein** that is required for **mineralization and maturation of the bone.**) This in turn enhances its **calcium binding properties**. It thus helps in the building and maintenance of bones.

**Deficiency of Vitamin K;**

The deficiency is rare though those with fat malabsorption syndrome e.g. celiac disease, cystic fibrosis e.t.c. or those on **prolonged use of antibiotics since it can destroy the intestinal bacteria** are at a higher risk.

People who take anti-coagulants and new born babies can also be at a risk of vitamin K deficiency. The deficiency is life threatening because a small wound can cause someone to bleed to death. Its deficiency thus causes **hemorrhage.**

**N/B** Since new borns lack the intestinal bacteria they are given injections of vitamin K at birth to prevent hemorrhage.

**Toxicity of Vitamin K**

Its toxicity is rare especially from food in that the body excretes vitamin K much more rapidly



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than other fat soluble Vitamins.

**RDA of Vitamin K**
60- 100µg daily.

**Food sources of Vitamin K**
- Intestinal bacteria
- Green leafy vegetables e.g. spinach and broccoli e.t.c.
- Vegetable oils e.g. soya bean oil, cotton seed oil e.t.c.
- Milk, liver and eggs.

**B. Water soluble vitamins;**
Water soluble vitamins are vitamins that are absorbed directly into the blood stream and they move freely through the watery environment and within cells. These vitamins when consumed in excess amounts are excreted through urine thus have no toxicities. (non-toxic). They must be consumed daily because there are no reserve deposits.
Water soluble vitamins are found in the watery portion of foods. Some of them are partly lost in cooking procedures and are susceptible to destruction by heat and alkalinity.
Water soluble vitamins include;
i) Vitamin B complex
ii) Vitamin C.

**1. Vitamin B complex;**
A number of substances have been identified and grouped together under the heading B complex. ‘ Each of the B vitamins is a separate vitamin in name, structure and function.
Vitamin B group consists of various vitamins which include **$B_1$, $B_2$, $B_3$, $B_5$, $B_6$, $B_7$, $B_9$,** and **$B_{12}$**.
Most of the B vitamins are synthesized in the GIT by microbial flora. They act as **coenzymes** when they collaborate with key cell enzymes in **performing energy metabolism** and **tissue building jobs.**
Milling process removes much of B vitamins from foods like cereals and other grains.
There are **8 B Vitamins** that are required by the human body that are divided into **three sub group.**
**Group A (classic deficiency/disease factors.**
i. Vitamin $B_1$- Thiamin
ii. Vitamin $B_2$- Riboflavin
iii. Vitamin $B_3$ – Niacin
**Group B (recently discovered co enzyme factors)**
i. Vitamin $B_6$ – Pyridoxine
ii. Pantothenic acid-vitamin $B_5$
iii. Biotin-vitamin $B_7$
**Group C ( blood forming factors)**



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i. Folic acid -$B_9$
ii. Vitamin $B_{12}$- cyanocobalamin

**1) Thiamin (Vitamin $B_1$)**

Thiamin is a water soluble vitamin that is also known as **aneurin**. It was isolated in 1926 from rice polishing by Jansen and Donath. It was subsequently synthesized in 1936 by R.R. Williams and is now available in the market in the form of Thiamin hydrochloride. The name thiamin is derived from its chemical ring like structure. It contains a sulphur containing ring (thiazole ring) and a nitrogen containing ring ( pyrimidine ring) attached to a carbon atom linked by a methyl bridge. Heat and alkali break the bond between the two rings.
(N/B-Draw the structure of Thiamin in your notebooks)

It exists mainly in various inter-convertible phosphorelated forms, chiefly as **thiamin pyrophosphate (TPP).** It is very soluble in water and is readily broken down by heat in neutral or alkaline solutions. Thiamin has a faint yeast like odour and a salty nut-like taste. It is fairly stable in mildly acidic solution.
Thiamin consumed in the diet is available both in free form and in bound form as thiamine pyrophosphate or as a protein phosphate. All the bound forms are spilt before being absorbed in the duodenum where acid medium activates its absorption.
Thiamin is not stored in the body thus daily intake is essential.
There are factors that increase the need for thiamin in the body which include;

i. Fever because of increased energy needs
ii. Increased muscular activity due increased energy needs
iii. Pregnancy and lactation
iv. Carbohydrates increased diet

N/B fats and protein intake spares thiamin. Any excess thiamin is excreted in urine.

**Functions of thiamin;**



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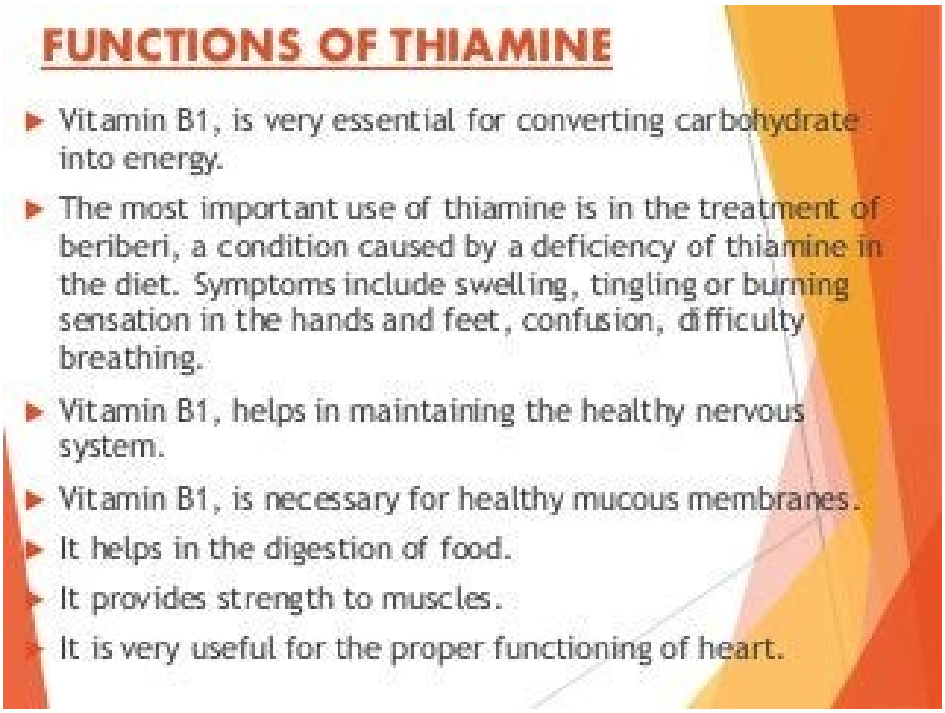


**Thiamin Deficiency;**

Thiamin deficiency is associated with heavy alcohol consumption particularly combined with reduced food intake. Alcohol interferes with the absorption of thiamin.

Thiamin deficiency is called **Beri beri** which is a **Philippino** word meaning **"I can't, I can't"** which refers to lack of neuromotor coordination in persons with the disease. Beri beri occurs in people whose major source of energy is polished rice. Deficiency of thiamin affect cardiovascular system, muscular and nervous system and the GIT system.

**Signs of thiamin deficiency;**

- ✓ Weakness
- ✓ Irritability
- ✓ Headache
- ✓ Fatigue
- ✓ Depression
- ✓ Muscle weakness
- ✓ Loss of appetite
- ✓ Nerve degeneration

When the symptoms of thiamin deficiency progress, they result in the following types of beri beri;

`i) **Dry beri beri;** this is mainly due to the central nervous system; the deficiency causes nerve degeneration, loss of nerve transmission leading to nervous tingling through-out the body,



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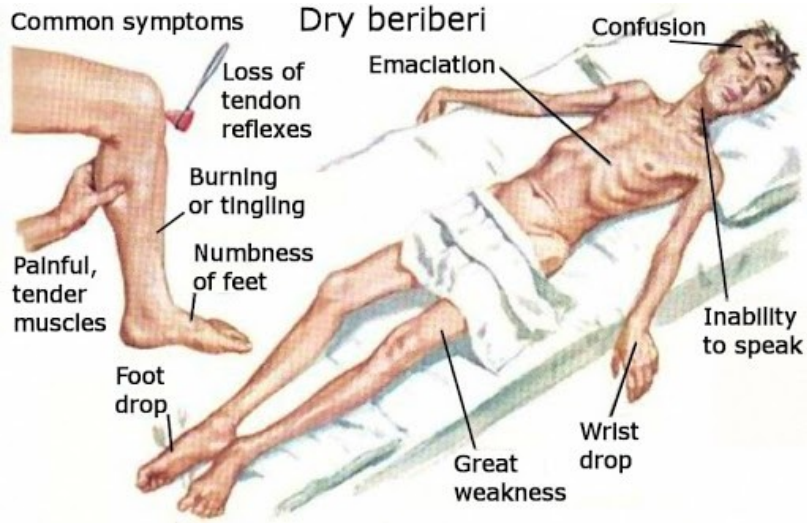
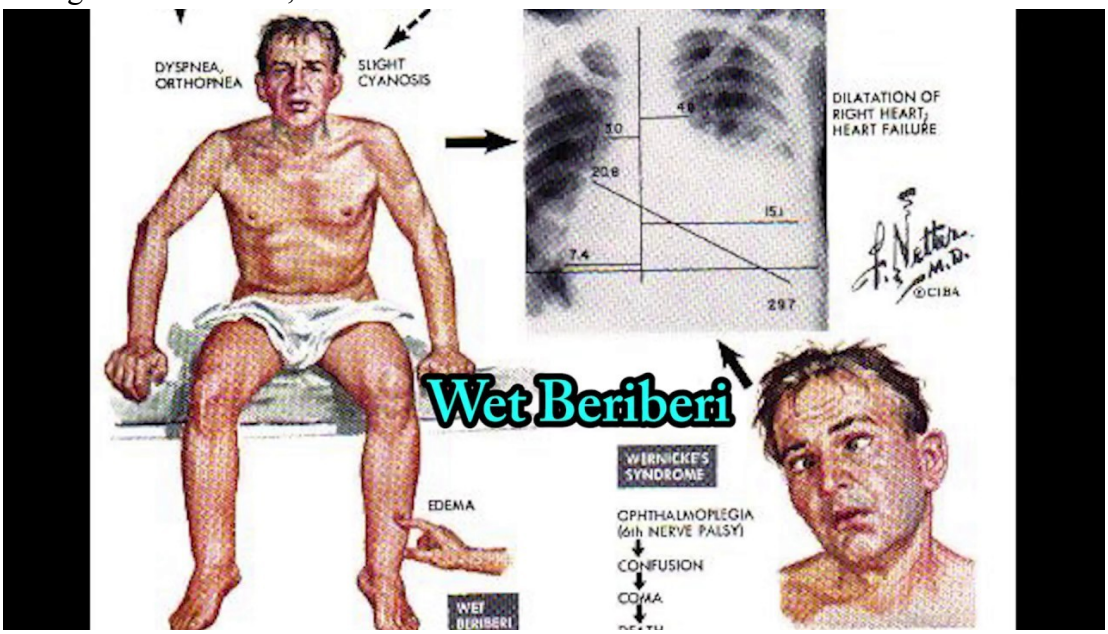
muscle wasting, poor arm and leg coordination and deep pain in calf muscle.




ii) **Wet beri beri**; is a deficiency that mainly affect the circulatory system (CVDs) it results in enlarged heart, heart failure and severe oedema.




iii) **Wernicke-korsakoff syndrome**; (**wernickes encephalopathy**). This condition is mainly due to alcohol induced malnutrition. Its symptoms are;

- ✓ mental confusion
- ✓ constant rapid eye movement (paralysis of the eye muscle)

iv) **Infantile beri beri**; - it occurs in infants between 2 to 5 months of age. The condition develops frighteningly and if not treated could lead to death. The affected infant develops **cyanosis** in which lack of oxygen or accumulation of *carbondioxide* in the blood causes the skin to **turn blue**, **tachycardia** and **a loud**, **piercing cry** that changes to **a thin, weak and**



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**almost inaudible one.** These symptoms are sometimes accompanied by vomiting and convulsions. Beri beri occur **more often in breast fed** than in bottle fed infants because a lactating mother's intake of thiamin may be too low to supply sufficient amount to enter her milk.

**RDA of thiamin;**

Daily requirements of B1 vary from 1.2-1.8mg/day. Its requirements depends on the size and energy use among individuals i.e. **0.5mg/1000kcals.**

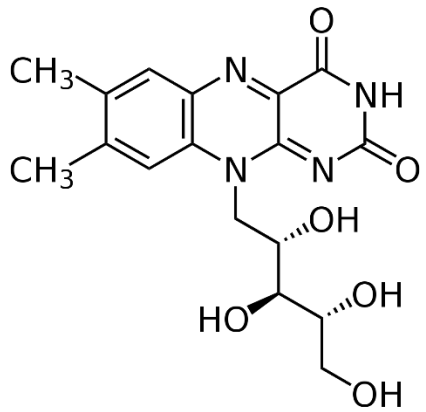
Its requirements increase with hypermetabolic state e.g. during fever, hyperthyroidism, intake of alcohols, surgery and pregnancy

**Food Sources of thiamin;**

- ✓ the richest sources- Pork and wheat
- ✓ Good sources are; cereals grains, legumes, sunflower seeds, yeast, water melon, nuts and organ meats e.g. liver.

**2. Vitamin $B_2$ (Riboflavin);**

It is also called **Lactoflavin.**




It is a yellow green (Latin word Flavus means yellow) fluorescent pigment containing the sugar ribose hence the name riboflavin. It has an Isoalloxazine ring and a ribitol in its structure. It is a vitamin that is resistant to acid, heat and oxidation but sensitive to light which can easily break it down.

Riboflavin is part of two coenzymes referred to as **Flavin mononucleotide (FMN)** and Flavin **adenine dinucleotide (FAD).** It is a vitamin that accepts and donates electrons and thus participates in several oxidation-reduction reactions.

It is less soluble in water than thiamin and more stable to heat especially in acid solution. When in solution, riboflavin is destroyed on exposure to sunlight e.g. prolonged exposure of milk to direct sunlight may decrease the riboflavin content considerably.



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**Functions of vitamin B2;**

1. FMN and FAD are **coenzymes** for a variety of enzyme reactions such as;
   i. In Flavoprotein of electron transport chain
   ii. Decarboxylation of pyruvate and ketoglutarate
   iii. Fatty acid acyl CoA dehydrogenase requires FAD in fatty acid oxidation
   iv. FAD is required in the production of pyridoxic acid from pyridoxal (Vitamin B6)
   v. The primary coenzyme form of Vitamin B6 i.e pyridoxal phosphate is dependent on FMN
   vi. FAD is required for the conversion of retinol (Vitamin A) to retinoic acid.
   vii. $FADH_2$ is required in the synthesis of the active form of folate (5-methyl THF)
   viii. FAD is required in the conversion of tryptophan to niacin
   ix. FAD is required in the reduction of the oxidized form of glutathione (GSSG) to reduced form (G-SH)
2. It is essential for **tissue health and growth of all animal and plant life including microorganisms.**
3. **It plays a role in maintaining the integrity of the mucous membrane.**

**Deficiency of Vitamin $B_2$;**

Biochemical signs of an inadequate $B_2$ status can appear after only a few days of a poor intake. The elderly and the adolescents are at a greater risk for riboflavin deficiency. $B_2$ deficiency interferes with iron handling capacity and contributes to **anemia** when iron intake is low.

Its deficiency is called **Ariboflavinosis** originally called **pellagra sin pellagra** (pellagra without pellagra) and it exhibits certain similarities to the $B_3$ deficiency pellagra. In humans, the classical syndrome affects;

1. The mouth; exhibiting the following symptoms;
   - ✓ **Sore throat**
   - ✓ **Cheilosis;** cracks at the corners of the lips



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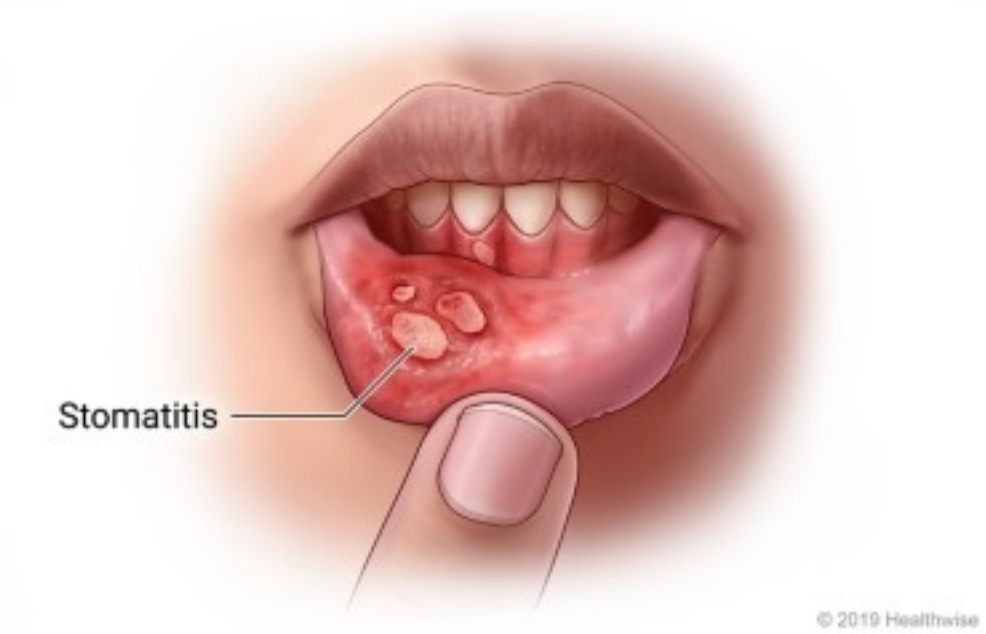
- ✓ **Glossitis;** soreness of the tongue (inflammation with a reddened surface)
- ✓ **Stomatitis;** inflammation of the mouth and lips that affect the mucous membrane lining the mouth and the lips.




2. The eyes; manifesting itself through; **photophobia** and **blood shot itchy watery eyes**
3. The skin; causing dermatitis (eczemia) which is the inflammation of the skin manifesting itself through itching, dryness, recurring rashes with redness, flaking, blistering, cracking, oozing or bleeding. It especially affect the scrotum, labia majora etc.
4. In blood; it decreases red blood cell count with normal size and haemoglobin content a condition known as **normochromic normocytic anaemia.** (circulating red blood cells



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(RBCs) are the same size (normocytic) and have a normal red color (normochromic)

5. In children it results in reduced growth i.e. retarded growth
6. Poor appetite
7. Vomiting.

N/B many of the symptoms may be related to the effect on metabolism of **folate** and **vitamin $B_6$.**

Cancer, heart disease and diabetes precipitate $B_2$ deficiency.

**RDA of vitamin B2;**

For adults is 1.2mg/day with calorie intake of >2000kcal
Adult men 1.3mg/day and women 1.1mg/day
Requirements increase with pregnancy and lactation
i.e. 1.4mg/day in pregnancy and 1.6mg/day in lactation
Infants' RDA is 0.3-0.4mg/day and children's is 0.6mg-0.9mg/day.

**Food sources;**

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3. **VITAMIN $B_3$ (NIACIN);**

This is a water soluble B vitamin that exists as **nicotinic acid** and **nicotinamide**. The body converts nicotinic acid to nicotinamide which is the major form of niacin in the blood. Nicotinamide is an amide derivative from nicotinic acid i.e. **pyridine 3- Carboxylic acid. (**the name **niacin** is used for **nicotinic acid** to avoid association with the **nicotine** of tobacco). Nicotinamide is a stable vitamin and is not easily destroyed by heat, acids, alkalis or by oxidation. It is a colourless water soluble solid.

Nicotinic acid and nicotinamide are both used by the body to form the coenzymes **nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP).** NAD can be converted to NADP by phosphorylation in the presence of the enzyme **$NAD^+$ kinase.** NADP and NAD are coenzymes for many **dehyrogenases participating** in many **hydrogen transfer processes**

NAD is important for the **catabolism** of fat, carbohydrates, protein and alcohol as well as cell signaling and DNA repair. NADP is involved in **anabolic reactions** e.g. fatty acid and cholesterol synthesis. This vitamin is related to the essential amino acid **tryptophan.** In that Trytophan is a precursor that can be converted by the body to niacin.

Niacin is a vitamin that **is most stable of all vitamins**. It forms a white powder when crystallized. Niacin is easily absorbed from the small intestines. Its precursor, tryptophan is converted to the vitamin in the presence of **pyridoxide ($B_6$)** and **vitamin $B_2$**

**Functions of vitamin $B_3$**

i. It is an important coenzyme that helps in **energy release from glucose, fat and alcohol in all body cells in form of NAD and NADP.** It works with vitamin $B_2$ and $B_1$
ii. NADP is an **important coenzymes** that helps in the synthesis of fatty acids and cholesterol in the body
iii. Niacin is used for **increasing the levels of high Density lipoprotein ( HDL) i.e. good cholesterol in the blood** which have been found to decrease the risk of cardiovascular diseases
iv. It is required for the normal functioning of the nervous system.
v. Niacin partner with riboflavin (vitamin $B_2$) In the cellular coenzyme system that converts protein to glucose, which is then oxidized to release controlled energy i.e. **gluconeogenesis**
vi. Niacin is a vitamin that is necessary for **growth, DNA repair, and production of steroid hormones in the adrenal gland.**
vii. It is used for **therapeutic purpose** because it has no side effects compared to other



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vitamins. It is used in managing cardiovascular diseases to lower the elevated levels of cholesterols.

**N/B** High doses of nicotinic acid (not nicotinamide) act as vasodilators and causes gastrointestinal distress, skin flushing (redness) and itching.

**Deficiency of Vitamin $B_3$**

- Lack of niacin affects the skin, gastro intestinal tract and the nervous system.
- The condition related to deficiency of vitamin $B_3$ is known as **pellagra**

- Deficiency of niacin is normally seen in condition of poverty, malnutrition and chronic alcoholism
- It also tends to occur in areas where people eat maize (the only cereal low in tryptophan and niacin) as a staple food
- Mild niacin deficiency has been shown to slow metabolism causing decreased tolerance to cold.
- Severe deficiency of niacin in the diet causes the disease called **pellagra** which is characterized by the **4Ds** i.e. **D**iarrhea, **D**ermatitis, **D**ementia, and **D**eath
- **Dermatitis** affects the skin especially the parts exposed to the sun e.g. a condition called " **Casal's necklace"** characterized by lesions on the lower neck; hyperpigmentation, thickening of the skin, itches, rough skin, burns, inflammation of the mouth and tongue.



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- **Dementia** is a nervous disorder characterized by dizziness, insomnia (difficulty in falling asleep/staying awake), amnesia (loss of memory), irritability, fear, depression, delirium (confusion), anxiety, forgetfulness, fatigue, disorientation, restlessness, apathy, poor concentration etc.
- **Diarrhea** is associated with digestive disturbances associated with lack of appetite
- If this condition is not corrected it can eventually lead to **Death**

**N/B** There is an hereditary nutritional disorders that result in niacin deficiency known as **Hartnup's diseases** which was first identified in the 1950s by the Hartnup's family in London it is due to deficit in the intestines and kidneys making it difficult for the body to break down and absorbs dietary tryptophan the precursor for niacin synthesis in the body which result in a condition similar to pellagra.

- Lack of niacin may also be observed in **pandemic deficiency disease** which is caused by lack of **five crucial vitamins** namely**; vitamin $B_3$, vitamin C, vitamin $B_1$, vitamin D and vitamin A** usually found in areas of widespread poverty and malnutrition.
- High energy requirements of the brain or high turnover rate of the gut and skin organs are usually the most susceptible to their deficiency.

**Food sources of Niacin**

i. Niacin can be synthesized from tryptophan in the body tissue if the diet is adequately supplied with proteins rich in tryptophan.
ii. Plant food e.g. groundnuts are the best sources, cereal grains (except maize), pulses,



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fruits, and vegetables (contain very small amounts).

iii. Brewer's yeast is a very concentrated sources of niacin it is sometimes added to recipe to increase niacin content of the diet.

iv. Animal sources e.g. liver (rich source) and meat.

**Recommended Daily Allowances (RDA) For Niacin**

- Since niacin is involved in the utilization of carbohydrates, the requirements of niacin are related to the total calories in the diet.
- The RDA is 6.6mg/1000 kilocalories.
- For children is 2-12 mg/day; women is 14mg/ day; men is 16mg/ day while pregnant and lactating women is 18mg/day

**N/B** Total upper intake levels (UL) is 35mg/day for women and men to avoid flashing.

**4. Pantothenic acid (Vitamin $B_5$);**

This is a vitamin derived from a Greek word **"pantheon"** meaning **all over or everywhere**. This shows that it is widely present in nature and is also freely synthesized by bacteria in the intestines. Hence there is never deficiency of this vitamin although it can occur in alcoholics.

It is a white crystalline compound, free acid and unstable viscous yellow oil which is stable in water. It is sensitive to acids, alkali and heat. It is absorbed readily in the intestines. It combines with phosphorous to form the active co-enzyme; **co-enzyme A** which has widespread metabolic presence throughout the body.

**Functions of B5;**

In form of **co-enzyme A** it has so many vital roles in the body e.g.

- ✓ It forms active acetate which is very important in the process of energy production in the cells from both fats and carbohydrates.
- ✓ The active form of acetate is also a precursor of cholesterol and other steroid hormones synthesis.
- ✓ It helps in the activation of succinic acid and glycine which is part of the primary step in the formation of heme in haemoglobin synthesis.
- ✓ **It combines with sulfonamide drugs to facilitate their excretion.**

**RDA OF $B_5$;**

RDA for this vitamin has not been stated because of easy availability though average intake is 10-20mg/day.

**Food sources;**

Poultry, organ meats, yeast, egg York, oats, fish, whole grains, potatoes and Broccoli.

**5. Vitamin $B_6$ (pyridoxine);**

This is a vitamin that has a pyridine ring in its structure. It has three basic forms;



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i. Pyridoxol
ii. Pyridoxamine
iii. Pyridoxine

All these are converted to **coenzyme pyridoxal phosphate** ($B_6$-$PO_4$) which is the most active form of this vitamin. It is water soluble, stable to heat and acids but sensitive to light and alkalis.

$B_6$ is easily absorbed from the duodenum. The active form of pyridoxal phosphate can be formed from any of the three isomer compounds. It is stored in all the tissues in the body.

**Functions of $B_6$;**

1. It is a **coenzyme in protein metabolism** when in its active form. It acts as a coenzyme in many amino acid reactions e.g;
   i. **Carboxylation** (chemical reaction in which a carboxylic acid group is produced by treating a substrate with carbon dioxide) which converts **glutamic acid** to **gamma-amino butyric acid** (GABA) and **tryptophan** to **serotonin** which are vital substances in the brain activities.
   ii. **Transamination** i.e transfer of one amino acid to the other releasing carbon residues for energy.
   iii. **Trans-sulfuration;** moves sulfur from sulfur-containing **methionine** to form other sulfur compounds like **cysteine.**
   iv. It controls the formation of **niacin** from **tryptophan**
   v. **Haemoglobin** synthesis by incorporating amino acids in 'heme' the essential non-protein core of haemoglobin.
   vi. **Amino acid transport**- it actively transports amino acids from the intestines into circulation and across cell walls into the body cells.
2. It is a coenzyme in carbohydrate and fat metabolism. It provides metabolites for energy producing fuels. It also **converts essential fatty acids** to **arachidonic acid.**
3. It plays a role in the formation of **antibodies.**

**Deficiency of vitamin $B_6$;**

- ✓ Anaemia since haemeglobin is not formed
- ✓ Central nervous system dysfunctions with hyperirritability
- ✓ Glossitis
- ✓ Cheilosis
- ✓ Seborrhoic dermatitis



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**RDA of B6;**
$B_6$ requirements may be related to protein intake. RDA for adults is 1.5-2.0mg/day
Infants 0.4mg/day
Pregnant and lactating mothers 2.5mg/day

**Food sources;**
Good sources; Pulses, wheat and other cereals, meat, liver and kidneys,
Fruits and vegetables are poor sources.

**6. Vitamin $B_7$ (Biotin);**
This is a vitamin that is found in minute traces in foods. It is found in free form or in protein bound form (**Biocytin).** It is stable to heat, light and acids.
It is an essential **sulfur-containing** vitamin that performs multiple tasks. It functions in partnership with **acetyl CoA** in reactions that **transfer carbon dioxide from one compound to another e.g;**

i. Initial steps in the synthesis of some fatty acids
ii. Conversion reactions in the synthesis of amino acids and in the production of antibodies thus help in increasing the effectiveness of the immune system.
iii. $CO_2$ fixation in the formation of purine

**Deficiency of B7;**
Its deficiency is very rare but patients on parenteral nutrition do show the deficiency. Long term alcohol abuse may also lead to biotin deficiency.
Pregnancy, DM and haemodialysis impairs uptake of the vitamin.

**RDA of B7;**
For adults is between 30-100mg/day.
**Food sources;**



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It is widely available in nature, e.g in corn and soya beans. It is also present in wheat though unavailable.
Good sources are; egg yolk (**though bound by avidin**), liver, kidneys, yeasts and tomatoes.
N/B breast milk has less than animal milk.

**7. Vitamin $B_9$ (Folate);**

Folic acid and related compounds are B vitamins that were discovered in 1941 as growth factor for bacteria. Its name is derived from a **Latin** word **Folium** which means **leaf** because it was first isolated from spinach leaves and is widely distributed in green leafy plants.
It is a **pteroyl-mono-glutamate acid**.

It is soluble in slightly alkaline or acidic solution. It is stable in neutral or alkaline solution especially in absence of air. It is water soluble and unstable in acid/ is sensitive to acid. It is relatively stable but cooking can result in losses as a high as 50% especially if cooking water is discarded.
There is a reduced form of folic acid called follinic acid which is made of 3 linked compounds called a pteridine group, para-amino benzoic acid and glutamic acid.
Small amounts of folic acid are synthesized by the intestinal microflora. About 25% folacin in free form in foods is completely absorbed. Polyglutamate form is only absorbed after it has been acted upon by the intestinal mucosal enzymes conjugase. By removing extra glutamate group conjugase release the absorbable folic acid. The **active form** of folic acid is **tetrahydrofolic acid (THF).**

**Functions of folate;**

1. **Basic coenzyme role**; it is an important agent in transferring a single carbon to a number of metabolite compounds e.g;
   i. **Purine;** nitrogen containing compounds essential to all living cells and involved in cell division and in transmission of inherited trait
   ii. **Thymine;** an essential compound forming an essential part in DNA, the material in the cell nucleus that controls and trans mits genetic materials.
   iii. **Haemoglobin;** folic acid is one of the important haemopoietic agents (formation of blood cells in the living body) necessary for proper blood formation.
2. It is involved in the production of **nucleic acid** together with **vitamin $B_{12}$.**

**Deficiency Folate;**

1. **Megaloblastic anaemia** which occurs in prolonged folic acid deficiency and is characterized by the formation of abnormal red blood cells.
2. **Celiac disease** a GIT problem characterized by inflammation of the mucosa, malabsorption



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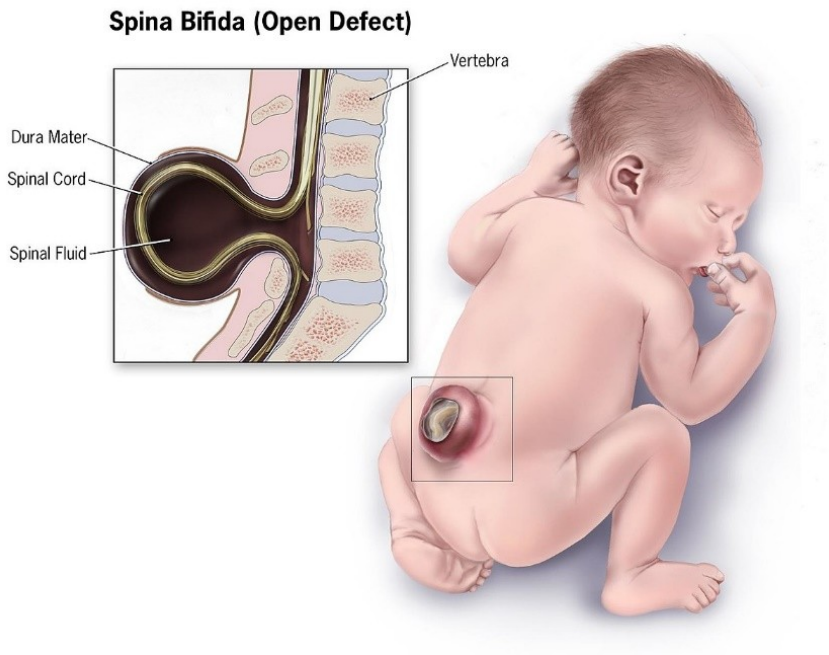
defects and general malnutrition.

3. It leads to **spina bifida** in children born to women who had folate deficiency during pregnancy. It is a developmental congenital disorder caused by incomplete closing of the embryonic neural tube. Some vertebrae overlying the spinal cord are not fully formed and remain unfused and open. If the opening is large enough it allows a portion of the spinal cord to protrude through the opening in the bones.




**RDA of folate;**

RDA for adults is 180-200mg/day. Pregnant women is 400mg/day

The additional needs of pregnancy may be met through folate supplements as it may be difficult to provide it in the diet (mandatory in MCH).

**Factors leading to high folic acid needs;**

i. **Chemotherapy** for cancer management e.g. amethopterin drug which acts as folic antagonists.
ii. **Stress and growth** increase folic acid requirements due to DNA synthesis

**Food sources folate;**

It is widely distributed in food e.g. green leafy vegetables, liver, kidneys, legumes and yeasts.

8. **VITAMIN $B_{12}$ ( Cyanocobalamin)**

This is the last member of the B vitamins discovered in 1948. It is a vitamin that contains cobalt and phosphorus and it is red in colour. It is the **only** B vitamin that is found in animals foods (i.e. it is the only B that does not occur in plants). Vitamin $B_{12}$ is not stable to heat or



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light hence processed foods are not rich sources. It is stored in active body tissues e.g. liver, heart, pancreases, testes, brain, blood and bone marrow.

It has a **corrin nucleus** which consists of a ring structure comprising four five membered rings containing nitrogen (i.e. **tetrapyrrole ring system).** In the active centre of the nucleus is a cobalt ion. A cyano group is attached to cobalt as an artifact of isolation.

It is a form of B vitamin which has some interdependent relationship with folate (i.e. each of these vitamins must have the other to be activated). Thus termed as **metabolic twins.**

Vitamin $B_{12}$ is a vitamin that was discovered through studies of pernicious anaemia a condition that begins with a **megaloblastic anemia** and leads to irreversible degeneration of the central nervous system (**pernicious anaemia**).

**Metabolism of Vitamin B $_{12}$:**

Vitamin B $_{12}$ can be absorbed from the intestines i.e. terminal ileum and it must be bound to a highly specific **glycoprotein** termed as the **intrinsic factor** which is secreted by **parietal cells** of the gastric mucosa.

Before absorption, vitamin $B_{12}$ must first be split (cleaved) from its protein source by hydrochloric acid and the action of digestive protease (pepsin) in both the stomach and the small intestines.

In man, the intrinsic factor may be lacking which leads to poor absorption of vitamin resulting in a condition known as **pernicious anemia.**

- ✓ People using antacid and acid blocking medication that inhibits absorption of vitamin $B_{12}$ are at risk of the deficiency.
- ✓ The elderly people who produce less acid are also at increase probability of $B_{12}$ deficiency.
- ✓ The deficiency is also seen in strict vegetarians.

**Functions of vitamin $B_{12}$**

i. Vitamin $B_{12}$ **promotes normal growth and development by aiding cell division.**
ii. It helps prevent certain types of nerve damage.
iii. It helps to treat **pernicious anemia.**
iv. It is essential **component of several coenzymes** needed in the synthesis of nucleic acids e.g. DNA, Isomerizes, dehydrates and enzymes involved in biosynthesis of methionine from homocysteine and metabolism of prep ionic acid to succinic acid i.e. conversion of methyl melonyl COA into succinyl COA.
v. It is used closely with **folate in the production of healthy red blood cells** and thus prevent anemia.
vi. Vitamin $B_{12}$ is required for the **normal functioning of cells especially those of the bone marrow, the nervous system and the gastrointestinal tract and formation of blood.**
vii. It is a vitamin that is required in **fatty acid synthesis and energy production.**

**Deficiency of vitamin $B_{12}$**

- ➢ Uncomplicated deficiency ( i.e. mild ) is characterized by symptoms such as:



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i. Sore tongue
ii. Weakness
iii. Loss of weight
iv. tingling of extremities
v. apathy( complete lack of emotion)
vi. Mental and nervous abnormalities.

- Where there is absence of intrinsic factor essential for absorption of the vitamin, pernicious anemia occurs in which there is degeneration of the spinal cord.
- When there is low levels of vitamin $B_{12}$ in the blood it results in an inability of new red blood cells (RBC) to develop normally resulting in **megaloblastic anemia.**
- This condition can be cured by injecting vitamin $B_{12}$ intramuscularly.

**Food sources vitamin $B_{12}$**

- Animal foods e.g. liver, organ meats, milk, muscle meat, fish and eggs.
- It is also biosynthesized by bacteria in the gastrointestinal tract. Its presence in animal foods is thought to be due to microbial origin.

**RDA for vitamin $B_{12}$** is 2.0mg/day

**2. Vitamin C (Ascorbic Acid)**

Vitamin C was isolated in the year 1932 by C.G. King. Its lack in human diet causes **scurvy.** It is the **most famous vitamin in fruits and vegetables**. Ascorbic acid closely resembles **glucose** in structure. It is a white, crystalline, odorless compound readily soluble in water. It has acidic and strong reducing properties.

```
  0===C
H0    C

H0    C

 H___ C

OH___C-H

      C-H2OH
```

**L-Ascorbic acid**

Vitamin C is heat stable in acid solution but is readily decomposed in the presence of alkali. The destruction of the vitamin is accelerated by exposure to light. It can also be destroyed by the action of oxygen and catalysts e.g. copper.

It is not stable in alkali pH therefore baking soda should not be added in food preparation to prevent vitamin C loss. Vitamin C is easily absorbed from the small intestines but lack of HCL or bleeding from GIT hampers its absorption.



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**Functions of vitamin C;**

i) Vitamin C is part of cementing material which holds the body cells firmly in place.
ii) It helps in maintaining strong tissues in general especially connective tissues e.g. bone, cartilages, dentin, skin and collagen.
iii) It is a vitamin that prevent scurvy by promoting collagen formation
iv) It helps in the formation strong capillary walls.
v) It is an important partner of protein for tissue synthesis.
vi) It is required in the growth stages of life.
vii) It found in the highly metabolically active tissues in the body e.g. brain, kidney, pancrease, liver, adrenal glands, thymus and the spleen.
viii) It helps the body to build resistance to infection
ix) It aids in the in the absorption of iron and makes it available for haemoglobin synthesis as well as formation of red blood cells.
x) Vitamin C is needed in wound healing, fractures, infection, fever and recovery.
xi) Vitamin C is an antioxidant that protects vitamins A, E, polyunsaturated fatty acids and iron from destruction by acting as a strong reducing agent which tie up free radicals and thus protect the body from their deleterious effects.
xii) Vitamin C helps in reducing the severity of cold symptoms because vitamin C reduces blood histamine levels.
xiii) It is required in the formation of some neurotransmitters, synthesis of thyroxine, metabolism of some amino acids and for normal immune functioning.

**Deficiency of vitamin C;**

If vitamin C is not present in sufficient quantities, the cementing material is not formed in adequate amounts. Tiny breaks occur in the walls of very small blood vessels and haemorrhage results. Tissues of the gums become tender and often bleed as condition known as **scurvy.**

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**Symptoms of vitamin C deficiency;**

i. Weakness
ii. Lack of appetite
iii. Scaly skin
iv. Spongy gums

v. Oedema
vi. Emaciation

vii. Haemorrhage in tissues e.g. bleeding gums
viii. Diarrhea
ix. Delayed wound healing
x. Frequent coughs and colds

Failure in collagen formation results in structural defects in bones, teeth, cartilages, connectives tissues and muscles.

Deficiency of vitamin C also leads to retarded growth in infants.

**Vitamin C requirements increases with;**



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i. Cigarette smoking because smoking increases oxidative stress and metabolic turnover (**the sum of all biochemical reactions**) of vitamin C
ii. In response to stresses e.g. fever, chronic illness, infection and wound healing
iii. During the use of drugs like aspirin which increases vitamin C excretion through urine.

**Food sources of vitamin C;**

Citrus fruits and juices such as oranges, lemon, tangerine.

Red and green peppers, broccoli, cauliflower,

Brussels tomatoes, straw berries and green leafy vegetables such as cabbage



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amaranthus,

sprouted pulses e.g. green grams which have proved to be of value during drought and famine conditions when other sources of vitamin C e.g fruits and vegetables are not available.

**RDA of Vitamin C;**

Vitamin C is not stored in the body. It has to be supplied daily. Even if excess in ingested, it is excreted in urine.

RDA of vitamin C increases with age from 20-40mg/day for children to 40mg/day for adults and 80mg/day for lactating mothers. The amount recommended is liberal.

**N/B** vitamin C is destroyed by high temperatures as such food should be cooked for the least period of time and kept covered. To curtail surface exposure to oxygen and wastage through fluids, the fruits and vegetables should not be cut into small pieces until the time of use. Use of pressure cooker saves a large amount of vitamin C.



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## UNIT SIX: MALNUTRITION

Malnutrition includes both **under nutrition -** acute malnutrition (i.e. wasting and/or nutritional oedema), chronic malnutrition (i.e. stunting), micronutrient malnutrition and inter-uterine growth restriction (i.e. poor nutrition in the womb) -and **over nutrition** (*overweight* and *obesity*[1]*)*.

Acute malnutrition or wasting (and / or oedema) occurs when an individual suffers from current, severe nutritional restrictions, a recent bout of illness, inappropriate childcare practices or, more often, a combination of these factors. It is characterized by extreme weight loss, resulting in low weight for height, and/or bilateral oedema, and, in its severe form, can lead to death. Acute malnutrition reduces resistance to disease and impairs a whole range of bodily functions. Acute malnutrition may affect infants, children and adults. It is more commonly a problem in children under-five and pregnant women, but nonetheless this varies and must be properly assessed in each context. Levels of acute malnutrition tend to be highest in children from 12 to 36 months of age when changes occur in the child's life such as rapid weaning due to the expected birth of a younger sibling or a shift from active breastfeeding to eating from a family plate, which may increase vulnerability.

The most visible consequences of acute malnutrition are weight loss (resulting in moderate or severe wasting) and/ or nutritional oedema (i.e. bilateral swelling of the lower limbs, upper limbs and, in more advanced cases, the face). Acute malnutrition is divided into two main categories of public health significance: **severe acute malnutrition** (SAM) and **moderate acute malnutrition** (MAM). MAM is characterised by moderate wasting. SAM is characterised by severe wasting and/or nutritional oedema.

The term ***global acute malnutrition*** (GAM) includes both SAM and MAM. Mild acute malnutrition also has consequences but is not widely used for assessment or programming purposes.

Acute malnutrition increases an individual's risk of dying because it compromises immunity and impairs a whole range of bodily functions. When food intake or utilisation (e.g. due to illness) is reduced, the body adapts by breaking down fat and muscle reserves to maintain essential functions, leading to wasting. The body also adapts by decreasing the activity of organs, cells and tissues, which increases vulnerability to disease and mortality. For reasons

[1]



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not completely understood, in some cases, these changes manifest as nutritional oedema. A 'vicious cycle' of disease and malnutrition is often observed once these adaptations commence.

### 8.1 Moderate Acute Malnutrition (MAM)

The burden of MAM (wasting) globally is considerable. Around 41 million children are moderately wasted worldwide and the management of MAM is finally becoming a public health priority, given this increase in mortality and the context of accelerated action towards achievement of Millennium Development Goals (MDGs) 3 and 4. Children with MAM have a greater risk of dying because of their increased vulnerability to infections as well as the risk of developing SAM, which is immediately life threatening.[2]

Some children with MAM will recover spontaneously without any specific external intervention; however, the proportion that will spontaneously recover and underlying reasons are not well documented.

### 8.2 Severe Acute Malnutrition (SAM)

Acute malnutrition is distinguished by its clinical characteristics of wasting and / or bilateral pitting oedema[3]

- **Marasmus** – severe wasting presenting as both moderate and severe acute malnutrition
- **Kwashiorkor** – bloated appearance due to water accumulation (nutritional bilateral pitting oedema)
- **Marasmic kwashiorkor**- is a condition which combines both manifestations.

#### 8.2.1 Kwashiorkor (Acute PEM)

It usually develops rapidly as a result of protein deficiency, or more commonly, it is precipitated by an illness such as measles or other infections. It occurs mainly in older infants and young children aged 1 to 3 years.

**Clinical signs**

- Poor growth characterized by loss of weight and growth failure (60 to 80% weight for age).

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- Oedema- accumulation of fluids in tissues causing swelling
- Wasting of muscles
- Mental changes characterized by apathy, misery, irritability and sadness.
- Hair changes- the hair becomes coarse, lack of luster, brownish, sparse and easily pulled out.
- Anaemia due to lack of protein to synthesize blood cells is worsened by iron deficiency, malaria and worm infestation.
- Skin changes/lesions- dark pigmentation in areas of friction i.e. patches like sun burns.
- Diarrhoea
- Moon face-cheeks appear swollen with fatty tissues or fluids.
- Poor appetite/ anorexia
- Hepatomegaly (enlarged fatty liver).

**8.2.2 Marasmus (Chronic PEM)**

Occurs commonly in children from 6 to 18 months of age. The deficiency is primarily due to lack of energy especially in starvation and is worsened by infections and parasitic diseases.

***Clinical signs***

- Growth failure (weight for age less than 60%)
- Severe muscle wasting
- Alert though with deep sunken eyes
- Good appetite (violent suckling of hands)
- The face is wrinkled and drawn in (look like an old man)
- Diarrhoea
- Dehydration.

**8.2.3 Marasmic kwashiorkor**

It is a mixed form of PEM, and manifests as oedema occurring in children who may or may not have other signs of kwashiorkor. It reflects as a deficiency of both energy and protein.

**8.2.3.1 Clinical signs**

- Flaky paint dermatitis
- Mental changes
- Hepatomegaly



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- Diarrhoea
- Dehydration
- Oedema

#### 8.2.3.2 Prevention of PEM

PEM can be prevented by:

a. Giving children adequate food that is energy and micronutrient dense.
b. Exclusively breastfeeding infants for the first six months and then introduction of nutritious complementary foods starting at six months with continued breastfeeding up to two years and beyond.
c. Preventing or treating infectious diseases promptly since they increase nutrient needs.
d. Deworming children to eradicate worm infestations which contribute to low nutrient utilization.
e. Enhancing proper sanitation when handling, preparing and feeding children to prevent contamination. The water used to prepare feeds should be clean.
f. Promoting good maternal nutrition since it contributes to the health outcome of the baby. Poor maternal nutritional status leads to delivering preterm and low birth weight babies who are at risk of developing PEM.
g. Promoting health education especially among the care givers of children.

## 8.3 Micronutrient Deficiencies

Micronutrients are minerals and vitamins that are needed in tiny quantitûies (and are therefore known as *micro*nutrients). Micronutrient deficiencies account for roughly 11% of the under-five death burden each year. It is now recognized that poor growth in under-fives results not only from a deficiency of protein and energy but also from an inadequate intake of vital minerals (e.g., zinc), vitamins, and essential fatty acids.

Vitamins are either water-soluble (e.g. the B vitamins and vitamin C) or fat-soluble (e.g. vitamins A, D, E and K). Essential minerals include iron, iodine, calcium, zinc, and selenium

### 8.3.1 Key micronutrient deficiencies

- Iron deficiency leads to iron deficiency anaemia
- Vitamin C deficiency leads to scurvy



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- Vitamin A deficiency leads to exophthalmia
- Niacin or Vitamin B3 deficiency leads to pellagra
- Iodine deficiency leads to goitre and cretinism (in infants born to iodine deficient mothers)
- Thiamin or B1 deficiency leads to beriberi
- Riboflavin deficiency leads to ariboflavinosis
- Vitamin D deficiency leads to rickets

The main cause of micronutrient malnutrition is usually an inadequate dietary intake of vitamins and/or minerals.

### 8.3.2 Causes of Malnutrition

UNICEF'S conceptual framework on the causes of malnutrition:

a. **Immediate causes**: Inadequate food intake and presence of disease
b. **Underlying causes**: inadequate household food security; inadequate maternal and child care; inadequate health services and unhealthy environment
c. **Basic causes**: resources and control; human; economic and organizational; political and ideological superstructure; potential resources; technology; environment; people

**Conceptual Framework of Malnutrition**



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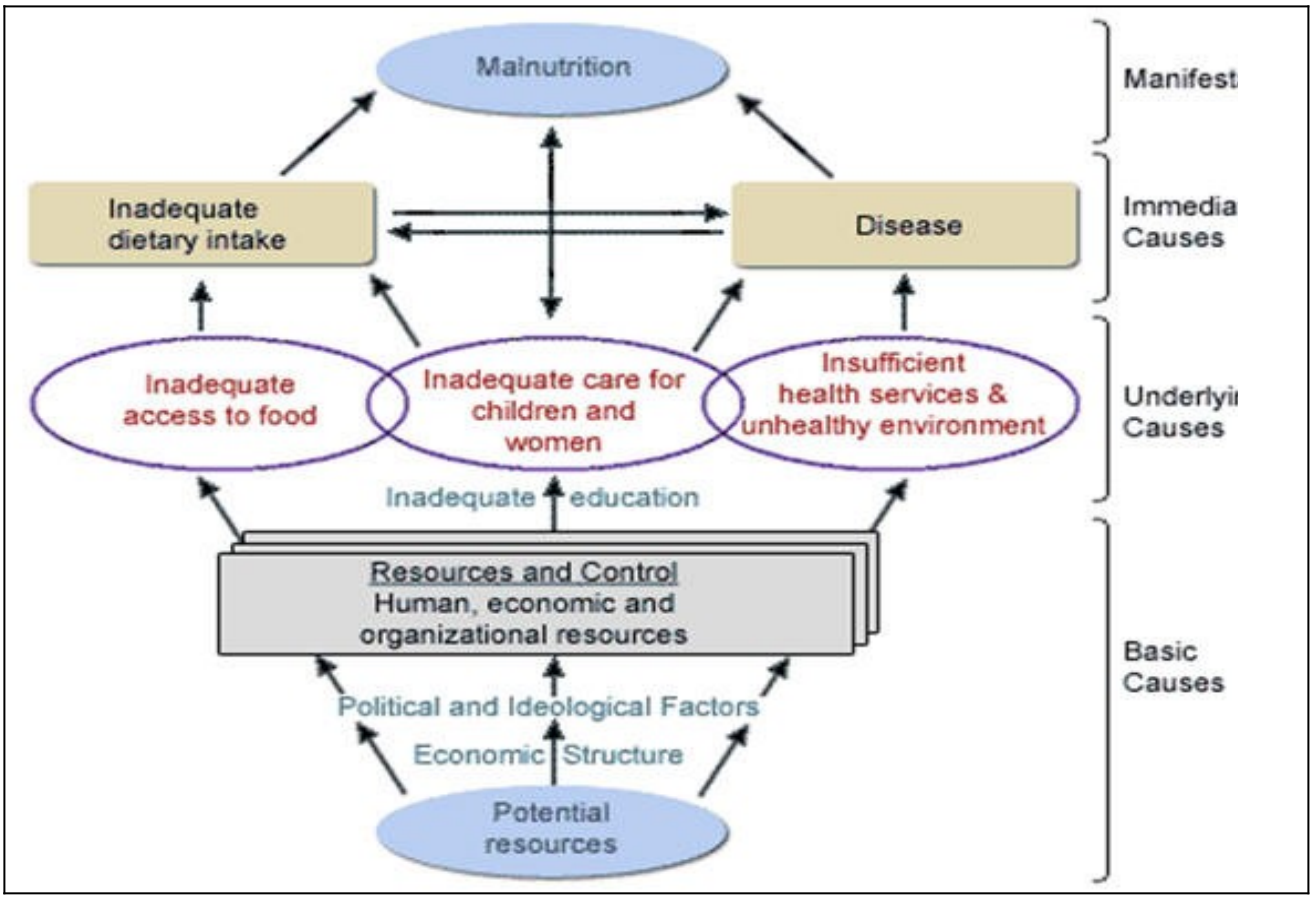


The basic causes of malnutrition are greatly influenced by:

- Political situations and economic systems that determine how resources are distributed
- Legal factors that determine how far the rights of women and girls are protected by law
- Cultural factors i.e. the ideologies and policies governing social sectors

To address basic causes of malnutrition, greater and better targeted resources and better collaboration are needed between sections of national governments and donors. United Nations Agencies, non-governmental organizations and investors. Above all, the people themselves must be a major part of the process.

**Solution Tree**



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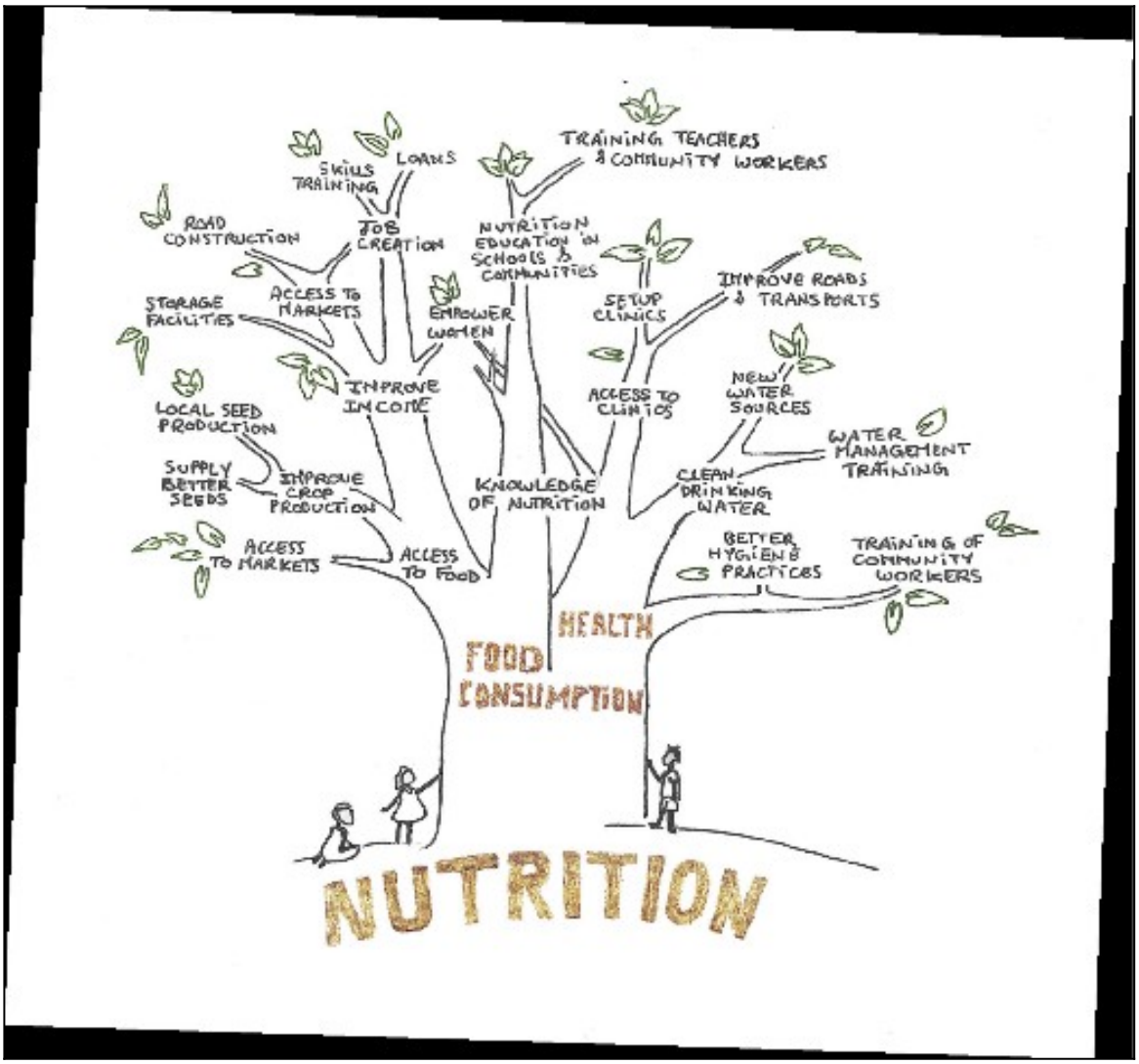


To address the problem of malnutrition, a multi-sectoral approach is used. Many sectors work together to address the different causes. The preparation of a solution tree will present "Reduction of malnutrition" as a common objective for an integrated approach addressing the various causes of malnutrition, bringing together different technical departments within an organization, or highlighting potential partnerships between organizations with complementary expertise. A health agency providing nutrition treatment, for example, could partner with a food security agency that will assist patients' families in diversifying their food production, income sources and food consumption, thus preventing relapses.



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